

SPECTRAL SEQUENCES AND APPLICATIONS

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1. TO READERS

1.1. About this lecture. These are lecture notes that I wrote for talks I gave in a seminar on [RB82] during Fall 2022. The goal of this lecture is to introduce applications of spectral sequences, and the most exciting applications are Serre's celebrated theorems about homotopy groups.

This lecture is divided into three parts:

1. In the **first part**, we establish the foundations of spectral sequence in a quick way, and use zig-zag to describe the differential maps d_r .
2. In the **second part**, we first introduce the spectral sequence that we are most concerned with, namely the Leray spectral sequence. Using this spectral sequence, we can compute (de Rham) cohomology groups of a (smooth) manifold, if it can be embedded into a fiber bundle and the cohomology of the other two spaces is well known.

The advantage of cohomology is that it admits a product structure, so we want to compute the cohomology ring structure of a given manifold using Leray spectral sequence. However, in this case, we need to be careful when computing the cohomology ring structure of the total space of a fiber bundle, since there is a so-called multiplicative extension problem, and the most typical example is $S^2 \rightarrow \mathbb{CP}^3 \rightarrow S^4$.

In order to consider torsion information of (co)homology groups, we give a quick review about singular (co)homology with different coefficients, establish the spectral sequence for them and then we prove the de Rham theorem using spectral sequence.

Using these tools, we give several interesting computations, such as cohomology ring structures of some Lie groups, and the path spaces of spheres.

3. In the **third part**, first we give a quick review of basic homotopy theory, such as homotopy exact sequence, Hurewicz theorem and the Bott periodicity theorem, and use these tools to compute the homotopy groups of Stiefel manifolds under a given dimension.

The final goal of this part is to compute homotopy groups of spheres and prove Serre's theorems about finitely generated and torsion property of homotopy groups of spheres. Serre's original ideas to compute homotopy groups of spheres are listed as follows:

- (a) Let π_q denote $\pi_q(X)$, where X is a topological space. If we want to compute π_q , just consider homology of $K(\pi_q, n)$, since Hurewicz theorem implies the n -th homology of $K(\pi_q, n)$ is exactly π_q .
- (b) If we can fit $K(\pi_q, n)$ into a fibration, maybe we can use Leray spectral sequence to compute its homology. Luckily, Postnikov approximation will give desired fibration.

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Part 1. Spectral Sequences

2. EXACT COUPLES

A simple way to construct a spectral sequence is through exact couples.

Definition 2.1 (exact couple). An exact couple is an exact sequence of abelian groups of the form

$$\begin{array}{ccc} A & \xrightarrow{i} & A \\ & \swarrow k & \searrow j \\ & B & \end{array}$$

where i, j and k are group homomorphisms.

From an exact couple, we can define a homomorphism $d : B \rightarrow B$ by $d = j \circ k$. Then $d^2 = 0$, so the homology group $H(B) = \ker d / \operatorname{im} d$ is well-defined.

Furthermore, from this exact couple, we can define a new exact couple, called derived couple,

$$\begin{array}{ccc} A' & \xrightarrow{i'} & A' \\ & \swarrow k' & \searrow j' \\ & B' & \end{array}$$

by making the following definitions.

1. $A' = i(A)$ and $B' = H(B)$;
2. i' is induced from i , that is $i'(ia) = i(ia)$;
3. For $a' = ia$ with $a \in A$, define $j'a' = [ja]$. To show j' is well defined, we need to check the following things
 - a. ja is a cycle. Indeed, $d(ja) = jkja = 0$;
 - b. The homology class $[ja]$ is independent of the choice of a . Indeed, if $a' = i\bar{a}$ for some other $\bar{a} \in A$. Then $a - \bar{a} = kb$ for some $b \in B$, since $a - \bar{a} \in \ker i = \operatorname{im} k$. Thus

$$ja - j\bar{a} = jkb = db$$

that is $[ja] = [j\bar{a}]$.

4. k' is induced from k . Let $[b] \in H(B)$, then $db = jkb = 0$ implies $kb \in \ker j = \operatorname{im} i$, so there exists $a \in A$ such that $kb = ia$. Define

$$k'[b] := kb \in i(A) = A'$$

Note that we also need to check k' is well-defined: take another $b' \in [b]$, that is $b' - b = db''$ for some $b'' \in B$. Then

$$kb' = kb + kdb'' = kb + kjkb'' = kb$$

As we have already defined these homomorphisms i', j' and k' , it suffices to check that the above diagram is an exact sequence. Let's check step by step:

1. $\text{im } j' = \ker k'$: Take $j'a' \in \text{im } j'$, then $k'j'a' = k'j'(ia) = k'[jia] = k jia = 0$; Conversely, if $[b] \in B'$ such that $k'[b] = kb = 0$, that is $b \in \ker k = \text{im } j$. So there exists $a \in A$ such that $b = ja$, so $[b] = [ja] = j'a'$, where $a' = ia$.
2. $\text{im } k' = \ker i'$: Take $k'[b] = kb \in \text{im } k'$, then $i'kb = ikb = 0$; Conversely, if $ia \in A'$ such that $i'ia = iia = 0$, so there exists $b \in B$ such that $ia = kb$. Furthermore, such b must be a cycle, since $jk b = jia = 0$. So $ia = kb = k'[b]$.
3. $\text{im } i' = \ker j'$: Take $iia \in \text{im } i'$, then $j'(iia) = [jia] = 0$; Conversely, if $ia \in A'$ such that $j'ia = [ja] = [0]$, that is there exists $b \in B$ such that $db = jkb = ja$, that is $a - kb \in \ker j = \text{im } i$. So there exists $a' \in A$ such that $a - kb = ia'$. So $a - ia' \in \text{im } k = \ker i$, that is $ia = iia'$. This completes the proof.

3. THE SPECTRAL SEQUENCE OF A FILTERED COMPLEX

In this section we fix a differential graded complex $K = \bigoplus_{k \in \mathbb{Z}} C^k$ with a differential operator $D : C^k \rightarrow C^{k+1}$.

Definition 3.1 (filtration). A sequence of subcomplexes

$$K = K_0 \supset K_1 \supset K_2 \supset K_3 \supset \dots$$

is called a filtration on K .

Notation 3.1. We usually extend the filtration to negative indices by defining $K_p = K$ for $p < 0$.

Definition 3.2 (filtered complex). A complex K with a filtration $\{K_p\}_{p \in \mathbb{Z}_{\geq 0}}$ is called a filtered complex and the associated graded complex is defined as

$$GK = \bigoplus_{p=0}^{\infty} K_p / K_{p+1}$$

Consider

$$A = \bigoplus_{p \in \mathbb{Z}} K_p$$

A is again a differential complex with operator D . Define $i : A \rightarrow A$ to be the inclusion $K_{p+1} \hookrightarrow K_p$ and define B to be the quotient, then we obtain a short sequence

$$0 \rightarrow A \xrightarrow{i} A \xrightarrow{j} B \rightarrow 0$$

and it induces a long exact sequence

$$\dots \rightarrow H^k(A) \xrightarrow{i_1} H^k(A) \xrightarrow{j_1} H^k(B) \xrightarrow{k_1} H^{k+1}(A) \rightarrow \dots$$

In other words, we can write it as an exact couple as follows

$$\begin{array}{ccc} A_1 & \xrightarrow{i} & A_1 \\ & \swarrow k_1 & \searrow j_1 \\ & B_1 & \end{array}$$

where $A_1 = H(A)$, $B_1 = H(B)$ and $i = i_1$. We suppress the subscript of i_1 to avoid cumbersome notation later. This exact couple gives rise to a sequence of exact couples:

$$\begin{array}{ccc} A_r & \xrightarrow{i} & A_r \\ & \swarrow k_r & \searrow j_r \\ & B_r & \end{array}$$

Example 3.1. Let's see a simple example: Consider a filtered complex that terminates after K_3 , that is

$$\dots = K_{-1} = K_0 \supset K_1 \supset K_2 \supset K_3 \supset 0$$

Then by definition, A_1 is the direct sum of all terms in the following sequence

$$\dots \xleftarrow{\cong} H(K) \xleftarrow{\cong} H(K) \xleftarrow{i} H(K_1) \xleftarrow{i} H(K_2) \xleftarrow{i} H(K_3) \leftarrow 0$$

And by definition of A_2 , it equals iA_1 , so it is the direct sum of all terms in the following sequence

$$\dots \xleftarrow{\cong} H(K) \xleftarrow{\cong} H(K) \supset iH(K_1) \xleftarrow{i} iH(K_2) \xleftarrow{i} iH(K_3) \leftarrow 0$$

Note that $iH(K_1) \subset H(K)$, and $i: H(K) \rightarrow H(K)$ is the identity map, thus $iiH(K_1) = iH(K_1)$. So A_3 is the direct sum of all terms in the following sequence

$$\dots \xleftarrow{\cong} H(K) \xleftarrow{\cong} H(K) \supset iH(K_1) \supset iiH(K_2) \xleftarrow{i} iiiH(K_3) \leftarrow 0$$

Similarly we have A_4 is the sum of

$$\dots \xleftarrow{\cong} H(K) \xleftarrow{\cong} H(K) \supset iH(K_1) \supset iiH(K_2) \supset iiiH(K_3) \supset 0$$

Since all terms appearing in A_4 are in $H(K)$, then i is identity on A_4 . So A 's are stationary after A_4 and we define

$$A_4 = A_5 = \dots = A_\infty$$

Furthermore, since $\ker\{i: A_4 \rightarrow A_5\} = \text{im } k_4$, thus $k_4 = 0$. Therefore after the fourth stage all the differentials of the exact couple are zero, since $d = jk$. So B 's are also stationary, that is

$$B_4 = B_5 = \dots = B_\infty$$

In the exact couple

$$\begin{array}{ccc} A_\infty & \xrightarrow{i_\infty} & A_\infty \\ & \nwarrow k_\infty=0 & \swarrow j_r \\ & B_\infty & \end{array}$$

A_∞ is the direct sum of groups

$$\dots \xleftarrow{\cong} H(K) \xleftarrow{\cong} H(K) \supset iH(K_1) \supset iiH(K_2) \supset iiiH(K_3) \supset 0$$

So if we let the above sequence be a filtration of $H(K)$, then B_∞ is the associated graded complex of the filtered complex $H(K)$.

Now let's come back to general case. The sequence of subcomplexes

$$\dots = K = K \supset K_1 \supset K_2 \supset K_3 \supset \dots$$

induces a sequence in cohomology

$$\dots \xleftarrow{\cong} H(K) \xleftarrow{\cong} H(K) \xleftarrow{i} H(K_1) \xleftarrow{i} H(K_2) \xleftarrow{i} H(K_3) \leftarrow \dots$$

Note that the maps i are of course no longer inclusions. Let F_p be the image of $H(K_p)$ in $H(K)$. For example, $F_3 = iiiH(K_3)$. There exists a sequence of inclusions

$$H(K) = F_0 \supset F_1 \supset F_2 \supset F_3 \supset \dots$$

making $H(K)$ into a filtered complex. This filtration is called the induced filtration on $H(K)$.

Definition 3.3 (length of filtration). A filtration K_p on the filtered complex K is said to have length l if $K_l \neq 0$ and $K_p = 0$ for $p > l$.

So as we can see from simple example we have computed, if the filtration of K has finite length, then A_r and B_r are stationary and the stationary value B_∞ is the associated graded complex $\bigoplus F_p/F_{p+1}$ of the filtered complex $H(K)$.

It is customary to write E_r for B_r , and there is a differential d_r on E_r such that $H_{d_r}(E_r) = E_{r+1}$, and that is the definition of a spectral sequence.

Definition 3.4 (spectral sequence). A sequence of differential complexes $\{E_r, d_r\}$ in which each E_r is the homology of its predecessor E_r is called a spectral sequence.

Definition 3.5 (convergence of spectral sequence). A spectral sequence $\{E_r, d_r\}$ is said to converge to some filtered group H , if E_∞ is equal to the associated graded group of H .

Let's summarize what we have done: For a differential complex K and a filtration $\{K_p\}$ of K , if the filtration is finite length, then the spectral sequence we obtained from this filtration will converge to $H(K)$.

However, it is quite a strong requirement for a filtration to be finite length. Suppose filtered complex $K = \bigoplus_n K^n$, then a filtration $\{K_p\}$ on K induces a filtration on K^n for each n , that is $K_p^n := K_p \cap K^n$. And we can prove the same result, only asking $\{K_p^n\}$ to be finite length for each n .

Theorem 3.1. Let $K = \bigoplus_n K^n$ be a graded filtered complex with filtration $\{K_p\}$ and let $H_D^*(K)$ be the cohomology of K with filtration given by $\{K_p\}$. Suppose that, for each n , the filtration $\{K_p^n\}$ has finite length. Then the short exact sequence of complex

$$0 \rightarrow \bigoplus K_{p+1} \rightarrow \bigoplus K_p \rightarrow \bigoplus K_p/K_{p+1} \rightarrow 0$$

induces a spectral sequence which converges to $H_D^*(K)$.

Proof. The idea here is that since this is a convergence between two graded groups, so it suffices to treat the convergence question one dimension at a time, then it is reduced to the ungraded situation.

Fix a number n and consider the n -th graded piece and let $\ell(n)$ be the length of $\{K_p^n\}_{p \in \mathbb{Z}}$, we have the following sequence

$$\dots \xleftarrow{\cong} H^n(K) \xleftarrow{i} H^n(K_1) \xleftarrow{i} H^n(K_2) \xleftarrow{i} \dots \xleftarrow{i} H^n(K_{\ell(n)}) \xleftarrow{i} 0 \xleftarrow{i} \dots$$

Use F_p^n to denote the image of $H^n(K_p)$ in $H^n(K)$. If $r \geq \ell(n) + 1$, then for all p

$$i^r H^n(K_p) = F_p^n$$

so we have

$$i: i^r H^n(K_{p+1}) \rightarrow i^r H^n(K_p)$$

is an inclusion, since both of them are in $H^n(K)$. By definition we have

$$A_r^n = \bigoplus_p i^r H^n(K_p)$$

and i_r sends $i^r H^n(K_{p+1})$ to $i^r H^n(K_p)$. It follows that

$$i_r: A_r^n \rightarrow A_r^n$$

is an inclusion thus $k_r : B_r^{n-1} \rightarrow A_r^n$ is the zero map. So we have $A_k^n = A_r^n$ and $B_k^{n-1} = B_r^{n-1}$ for all $k \geq r$, that is $A_\infty^n = A_r^n = \bigoplus_p F_p^n$ and $B_\infty^n = B_r^n = \bigoplus_p F_p^n / F_{p+1}^n$. Thus

$$B_\infty = \bigoplus_n B_\infty^n = \bigoplus_{n,p} F_p^n / F_{p+1}^n = \bigoplus_p F_p / F_{p+1}$$

which is the associated graded complex of $H_D^*(K)$, as desired. $\square$

4. THE SPECTRAL SEQUENCE OF A DOUBLE COMPLEX

4.1. Basic setting. Now for a double complex $K = \bigoplus_{p,q \geq 0} K^{p,q}$ with differential d and δ , we can make it into a complex, called the total complex with differential D by

$$K = \bigoplus_{k=0}^{\infty} C^k$$

where $C^k = \bigoplus_{p+q=k} K^{p,q}$ and $D = \delta + (-1)^p d = \delta + D''$. There is a natural filtration on K as follows

$$K_p = \bigoplus_{i \geq p, q \geq 0} K^{i,q}$$

The direct sum $A = \bigoplus_{p \geq 0} K_p$ is also a double complex, and we can also make it into a single complex $A = \bigoplus_{k \geq 0} A^k$ by summing the bidegrees.

Note that

$$A^k = \bigoplus_p A^k \cap K_p$$

and inclusion $i: A^k \rightarrow A^k$ is given by

$$i: A^k \cap K_{p+1} \rightarrow A^k \cap K_p$$

This gives an inclusion $i: A \rightarrow A$ and the quotient is denoted by B , where B is also a double complex, we can also make it into a single complex $B = \bigoplus_{k \geq 0} B^k$ by summing the bidegrees. We can write this short exact sequence as follows

$$0 \rightarrow \bigoplus_{k,p} A^k \cap K_p \rightarrow \bigoplus_{k,p} A^k \cap K_p \rightarrow \bigoplus_{k,p} B^k \cap (K_p/K_{p+1}) \rightarrow 0$$

where the differentials of these complexes are listed as follows:

1. A inherits the differential operator $D = \delta + (-1)^p d$ from K ;
2. $B = \bigoplus K_p/K_{p+1}$ also inherits the differential operator D , but D on B is just $(-1)^p d$, since any element in K_p is mapped into K_{p+1} by δ . Therefore

$$E_1 = H_D(B) = H_d(K)$$

Remark 4.1. From above section, we obtain a spectral sequence which converges to $H_D(K)$, since our filtration is finite on each degree n . However, we want to show a more refined theorem, since in this case our complex comes from a double complex, which has a more subtle structure. In order to do this, we need to compute the explicit formula of d_r .

Notation 4.1. We will denote the class of b in E_r , if it is well-defined, by $[b]_r$.

4.2. Explicit formula of d_r .

4.2.1. *Case of d_1 .* Note that

$$B^k = \bigoplus_p B^k \cap (K_p/K_{p+1})$$

So if we want to compute $k_1: H^k(B) \rightarrow H^{k+1}(A)$, it suffices to compute

$$k_1: H^k(B) \cap (K_p/K_{p+1}) \rightarrow H^{k+1}(A) \cap K_{p+1}$$

for each p .

Remark 4.2 (characterization of elements in E_1). Any element $[b]_1 \in H^k(B) \cap (K_p/K_{p+1})$ is $b + K_{p+1} \in B^k \cap (K_p/K_{p+1})$ such that $b \in K^{p,k-p}$ and $db = 0$. So you can regard $E_1^{p,q}$ as $H_d^{p,q}(K)$.

Now we fix p and consider

$$\begin{array}{ccccccc}
 & & \uparrow & & \uparrow & & \uparrow \\
 0 & \longrightarrow & A^{k+1} \cap K_{p+1} & \longrightarrow & A^{k+1} \cap K_p & \longrightarrow & B^{k+1} \cap K_p/K_{p+1} \longrightarrow 0 \\
 & & \uparrow D & & \uparrow D & & \uparrow d \\
 0 & \longrightarrow & A^k \cap K_{p+1} & \longrightarrow & A^k \cap K_p & \longrightarrow & B^k \cap K_p/K_{p+1} \longrightarrow 0 \\
 & & \uparrow & & \uparrow & & \uparrow
 \end{array}$$

In order to get $k_1[b]_1$, where $[b]_1 \in E_1^{p,k-p}$, we need to chase the diagram as follows

1. Choose $b \in A^k \cap K_p$ to represent $[b]_1$ ¹;
2. $Db = \delta b + (-1)^p db = \delta b \in A^{k+1} \cap K_p$, since $db = 0$;
3. Take inverse of $\delta b \in A^{k+1} \cap K_p$ under i , we obtain $\delta b \in A^{k+1} \cap K_{p+1}$.

Thus $k_1[b]_1 = [\delta b]_1 \in H^{k+1}(A) \cap K_{p+1}$. By definition of d_1 we can see

$$\begin{aligned}
 d_1 : H^k(B) \cap (K_p/K_{p+1}) &\rightarrow H^{k+1}(B) \cap (K_{p+1}/K_{p+2}) \\
 [b]_1 &\mapsto [\delta b]_1
 \end{aligned}$$

By characterization of elements in E_1 , we can regard $d_1[b]_1$ as $\delta b \in K^{p+1,k-p}$ with $d(\delta b) = 0$, and $[\delta b]_1 = 0 \in E_1$ is equivalent to saying there exists $c \in K^{p+1,k-p-1}$ such that $\delta b = -D''c$.

Remark 4.3 (characterization of elements in E_2). For an element of $[b]_2 \in E_2$, it can be represented by an element $b \in K$ with a zig-zag of length 2

$$\begin{array}{ccc}
 & 0 & \\
 & \uparrow d & \\
 b & \xrightarrow{\delta} & \delta b \\
 & \uparrow D'' & \\
 & c &
 \end{array}$$

In other words, $E_2 = H_\delta H_d(K)$.

For $[b]_2 \in E_2^{p,q}$, by definition of derived couple, we have

$$d_2[b]_2 = j_2 k_2 [b]_2 = j_2 [k_1 [b]_1]_2$$

In order to compute $j_2 [k_1 [b]_1]_2$, we need to find $a \in K$ such that $k_1 [b]_1 = i[a]_1$, then $j_2 [k_1 [b]_1]_2 = [j_1 a]_2$. Since $k_1 [b]_1 \in A^{k+1} \cap K_{p+1}$, we have $a \in A^{k+1} \cap K_{p+2}$.

¹It is clear the choice is not unique, any element taking form $b + c$, where $c \in A^k \cap K_{p+1}$ also can represent $b + K_{p+1}$.

To find such a we use not b but $b + c$ in $A^k \cap K_p$ to represent $[b]_1$, that's possible since b and $b + c$ have the same image under the projection $K_p \rightarrow K_p/K_{p+1}$, since $c \in A^k \cap K_{p+1}$. Then

$$k_1[b]_1 = D(b + c) = \delta b + Dc = \delta b + \delta c + D''c = i(\delta c) \in A^{k+1} \cap K_{p+1}$$

where $\delta c \in A^{k+1} \cap K_{p+2}$. So

$$d_2[b]_2 = [\delta c]_2$$

Thus differential d_2 is given by the delta of the tail of the zig-zag which extends b . By characterization of E_2 , you can regard it as an element in $H_\delta H_d(K)$. Now let us check well-definedness:

1. $\delta c \in H_\delta H_d(K)$: $\delta(\delta c) = 0$ is clear; $d\delta c = \delta dc = (-1)^p \delta \delta b = 0$, since $(-1)^p dc = \delta b$.
2. $d_2[b]_2$ is independent of the choice of c : Any two possible choices c and c' differ by an element lying in $\ker d$. Assume $c' = c + x$ where $x \in \ker d$, then it suffices to show $[\delta x]_2 = 0$, and that's tautological.

Remark 4.4 (characterization of elements in E_3). For an element of $[b]_3 \in E_3$, it can be represented by an element $b \in K$ with a zig-zag of length 3

$$\begin{array}{c}
 0 \\
 \uparrow d \\
 b \xrightarrow{\delta} \delta b \\
 \quad \quad \quad \uparrow D'' \\
 \quad \quad \quad c_1 \xrightarrow{\delta} \delta c_1 \\
 \quad \quad \quad \quad \quad \quad \uparrow D'' \\
 \quad \quad \quad \quad \quad \quad c_2
 \end{array}$$

Notation 4.2. We say that an element b in K lives to E_r if it represents a cohomology class in E_r , or equivalently, b is a cocycle in $E_1, E_2, \dots, E_{r-1}$. We have already seen that there is a zig-zag description for d_1 and d_2 .

Remark 4.5 (characterization of elements in E_r). Generally, an element $b \in K$ lives to E_r if it can be extended to a zig-zag of length r

$$\begin{array}{c}
 0 \\
 \uparrow d \\
 b \xrightarrow{\delta} \delta b \\
 \quad \quad \quad \uparrow D'' \\
 \quad \quad \quad c_1 \searrow \dots \\
 \quad \quad \quad \quad \quad \quad c_{r-2} \xrightarrow{\delta} \delta c_{r-2} \\
 \quad \quad \quad \quad \quad \quad \quad \quad \quad \uparrow D'' \\
 \quad \quad \quad \quad \quad \quad \quad \quad \quad c_{r-1}
 \end{array}$$

The differential d_r on E_r is given by δ of the tail of zig-zag:

$$d_r[b]_r = [\delta c_{r-1}]_r$$

Thus the bidegrees (p, q) of the double complex persist in the spectral sequence

$$E_r = \bigoplus_{p,q} E_r^{p,q}$$

and d_r shifts the bidegrees by $(r, -r+1)$.

$$d_r : E_r^{p,q} \rightarrow E_r^{p+r, q-r+1}$$

The filtration on $H(K)$

$$H(K) = F_0 \supset F_1 \supset F_2 \supset \dots$$

induces a filtration on each component $H^n(K)$ as follows

$$H^n(K) = (F_0 H^n) \supset \underbrace{(F_1 H^n)}_{E_\infty^{0,n}} \supset \underbrace{(F_2 H^n)}_{E_\infty^{1,n-1}} \supset \dots \supset (F_n H^n) \supset \underbrace{0}_{E_\infty^{n,0}}$$

where $F_i H^n := F_i \cap H^n(K)$. In summary, we have proven the following refinement:

Theorem 4.1. Given a double complex $K = \bigoplus K^{p,q}$ there is a spectral sequence $\{E_r, d_r\}$ converging to the total cohomology $H_D(K)$ such that E_r has a bigrading with

$$d : E_r^{p,q} \rightarrow E_r^{p+r, q-r+1}$$

and

$$\begin{aligned} E_1^{p,q} &= H_d^{p,q}(K) \\ E_2^{p,q} &= H_\delta^{p,q} H_d(K) \end{aligned}$$

Furthermore, the associated graded complex of the total cohomology is given by

$$GH_D^n(K) = \bigoplus_{p+q=n} E_\infty^{p,q}(K)$$

Remark 4.6. There is another filtration, that is $K_q = \bigoplus_{j \geq q, p \geq 0} K^{p,j}$. This gives a second spectral sequence $\{E'_r, d'_r\}$ converging to the total cohomology $H_D(K)$, but with

$$\begin{aligned} E'_1 &= H_\delta(K) \\ E'_2 &= H_d H_\delta(K) \end{aligned}$$

and

$$d'_r : E_r'^{p,q} \rightarrow E_r'^{p-r+1, q+r}$$

Example 4.1 (Revisit generalized Mayer-Vietoris principle). Given a smooth manifold M and an open covering $\mathcal{U}$ of it, consider the double complex $C^*(\mathcal{U}, \Omega^*)$, then there is only one column in E'_1 -page, therefore the E'_2 -page degenerates, which implies generalized Mayer-Vietoris principle. Furthermore, if we take good cover, the E_2 -page also degenerates, which implies

$$H_{dR}^*(M) \cong H^*(\mathcal{U}, \mathbb{R})$$

4.3. Additive extension problem. Since the dimension is the only invariant of a vector space, the associated graded vector space GV of a filtered vector space V is isomorphic to V itself. In particular, if a double complex K is a vector space, then

$$H_D^n(K) \cong GH_D^n(K) \cong \bigoplus_{p+q=n} E_\infty^{p,q}$$

However, the same thing fails in the realm of abelian groups. For example, consider filtered groups $\mathbb{Z}_2 \oplus \mathbb{Z}_2$ and $\mathbb{Z}_4$, which are filtered by

$$\mathbb{Z}_2 \subset \mathbb{Z}_2 \oplus \mathbb{Z}_2$$

and

$$\mathbb{Z}_2 \subset \mathbb{Z}_4$$

respectively. Thus they have isomorphic associated graded groups, but $\mathbb{Z}_2 \oplus \mathbb{Z}_2$ is not isomorphic to $\mathbb{Z}_4$. In other words, in a short exact sequence of abelian groups

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

A and C do not determine B uniquely. The ambiguity is called the (additive) extension problem.

Proposition 4.1. In a short exact sequence of abelian groups

$$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$$

if C is free, then there exists a homomorphism $s : C \rightarrow B$ such that $g \circ s$ is the identity on C .

Proof. Since C is free, it suffices to define a suitable s on the generators $\{c_i\}$ of C and it automatically extends to C linearly. Take c_i and choose any preimage of c_i , denoted by b_i , then s is defined by $c_i \mapsto b_i$. Clearly $s \circ g$ is the identity on C , but note that such s is not unique. $\square$

Corollary 4.1. Under the hypothesis of above proposition,

1. The map $(f, s) : A \oplus C \rightarrow B$ is an isomorphism;
2. For any abelian group G the induced sequence

$$0 \rightarrow \text{Hom}(C, G) \rightarrow \text{Hom}(B, G) \rightarrow \text{Hom}(A, G) \rightarrow 0$$

is exact;

3. For any abelian group G the sequence

$$0 \rightarrow A \otimes G \rightarrow B \otimes G \rightarrow C \otimes G \rightarrow 0$$

is exact.

Proof. For (1). Since (f, s) is a group homomorphism, it suffices to check that it is both injective and surjective. It is easy to see (f, s) is injective, since f and s are injective; For $b \in B$, if $b \in \text{im } f$, that is $b = f(a)$ for some $a \in A$, then $(a, 0)$ is mapped to b . If $b \notin \text{im } f = \ker g$, then consider $g(b) \in C$. Although $sg(b)$ may not equal to b , we have $sg(b) - b \in \ker g = \text{im } f$, so there exists $a \in A$ such that $f(a) + sg(b) = b$, this completes the proof of surjectivity.

For (2). Since it is well known $\text{Hom}(-, G)$ is a left exact functor, then it suffices to show $\text{Hom}(B, G) \rightarrow \text{Hom}(A, G)$ is surjective. Take any $k : A \rightarrow G$, then consider the composition of following maps

$$B \xrightarrow{(f,s)^{-1}} A \oplus C \xrightarrow{p_1} A \xrightarrow{k} G$$

it is a map in $\text{Hom}(B, G)$ such that it extends k .

For (3). Since it is well known $-\otimes G$ is a right exact functor, then it suffices to show $A \otimes G \rightarrow B \otimes G$ is injective, and the proof is quite similar as above. $\square$

Remark 4.7. According to standard facts in homological algebra, there are the following exact sequences

1.

$$0 \rightarrow \text{Hom}(C, G) \rightarrow \text{Hom}(B, G) \rightarrow \text{Hom}(A, G) \rightarrow \text{Ext}(C, G) \rightarrow \dots$$

2.

$$0 \rightarrow A \otimes G \rightarrow B \otimes G \rightarrow C \otimes G \rightarrow \text{Tor}(A, G) \rightarrow \dots$$

If G is an abelian group, then $\text{Ext}(-, G) = \text{Tor}(-, G) = 0$, which yields the desired results.

Part 2. Applications to cohomology theory

5. LERAY SPECTRAL SEQUENCE

Now let's focus on a special spectral sequence we are concerned about, that is Leray spectral sequence.

5.1. Basic setting. Let $\pi: E \rightarrow M$ be a fiber bundle with fiber F over a manifold M . Given a good cover $\mathfrak{U}$ of M , $\pi^{-1}\mathfrak{U}$ is a cover on E and we can form the double complex

$$K = C^*(\pi^{-1}\mathfrak{U}, \Omega^*)$$

with E_1 -page and E_2 -page as follows

$$E_1^{p,q} = H_d^{p,q}(K) = \prod_{\alpha_0 < \dots < \alpha_p} H^q(\pi^{-1}U_{\alpha_0 \dots \alpha_p}) = C^p(\mathfrak{U}, \mathcal{H}^q)$$

$$E_2^{p,q} = H_\delta^p(\mathfrak{U}, \mathcal{H}^q)$$

where $\mathcal{H}^q$ is a locally constant presheaf $U \mapsto H^q(\pi^{-1}U)$ on M . Furthermore, if M is simply-connected, then there is no monodromy, which implies $\mathcal{H}^q$ is a constant sheaf $\underbrace{\mathbb{R} \oplus \dots \oplus \mathbb{R}}_{\dim H^q(F)}$, thus

$$E_2^{p,q} = H^p(M) \otimes H^q(F)$$

By theorem 4.1 the spectral sequence of K converges to $H_D^*(K)$, which is equal to $H^*(E)$ by generalized Mayer-Vietoris principle, since $\pi^{-1}\mathfrak{U}$ is a cover of E .

Example 5.1 (orientability and the Euler class of sphere bundle). Let $\pi: E \rightarrow M$ be a S^n -bundle over a manifold M and let $\mathfrak{U}$ be a good cover of M . Then the E_2 -page of Leray spectral sequence is

$$E_2^{p,q} = H^p(\mathfrak{U}, \mathcal{H}^q(S^n))$$

Since only the 0-th and n -th cohomology groups of S^n are nonzero, there are only two nonzero rows on the E_2 -page. Thus $d_2 = \dots = d_n = 0$, and

$$E_{n+1} = E_2 = H_\delta H_d(K) = H^*(\mathfrak{U}, \mathcal{H}^*(S^n)).$$

Let $\sigma \in E_1^{0,n}$ be a collection of local angular forms. Then $d_1\sigma = 0$ precisely when the orientation local system is trivial. If E is oriented, σ therefore survives to E_{n+1} .

Up to sign, the only possible differential is

$$d_{n+1}\sigma \in E_{n+1}^{n+1,0} \cong H^{n+1}(M),$$

and this transgression is the Euler class $e(E)$. Thus a normalized global angular form exists if and only if $e(E) = 0$.

Example 5.2 (orientability of simply-connected manifold). Let M be a simply-connected manifold of dimension n and $S(T_M)$ is the S^{n-1} -sphere bundle of its tangent bundle. $H^1(M) = 0$ since M is simply-connected, thus let $\sigma \in E_1^{0,n-1}$ be the local angular forms on $S(T_M)$, we must have $d_1\sigma = 0$, since $E_2^{1,n-1} =$

$H^1(M) \otimes H^{n-1}(S^{n-1})$, thus $S(T_M)$ is orientable, that is T_M is orientable, which implies M is orientable.

Example 5.3 (The cohomology group of $\mathbb{CP}^2$). Consider the Hopf fibration of $\mathbb{CP}^2$, that is

$$\begin{array}{ccc} S^1 & \longrightarrow & S^5 \\ & & \downarrow \\ & & \mathbb{CP}^2 \end{array}$$

Since $\mathbb{CP}^2$ is simply-connected

$$E_2^{p,q} = H^p(\mathbb{CP}^2) \otimes H^q(S^1)$$

that is E_2 -page looks like

$$\begin{array}{c|ccccccccc} 1 & \mathbb{R} & A & B & C & D & & & & \\ & \searrow & \searrow & \searrow & \searrow & \searrow & & & & \\ 0 & \mathbb{R} & A & B & C & D & 0 & 0 & & \\ \hline & & 0 & 1 & 2 & 3 & 4 & 5 & 6 & \end{array}$$

Since d_3 moves down two rows, $d_3 = 0$, and similarly all later differentials vanish. Thus the spectral sequence degenerates at the E_3 -page. Its E_∞ -page is the associated graded vector space of $H^*(S^5)$, so it looks like

$$\begin{array}{c|cccccc} 1 & & & & & & \mathbb{R} \\ & & & & & & \\ 0 & \mathbb{R} & & & & & \\ \hline & & 0 & 1 & 2 & 3 & 4 \end{array}$$

which implies

$$0 \rightarrow A, \quad \mathbb{R} \rightarrow B, \quad A \rightarrow C, \quad B \rightarrow D, \quad C \rightarrow 0$$

are isomorphisms. Thus

$$H^q(\mathbb{CP}^2) = \begin{cases} \mathbb{R} & q = 0, 2, 4 \\ 0 & \text{otherwise} \end{cases}$$

Remark 5.1. By the same argument one can compute cohomology group of $\mathbb{CP}^n$.

5.2. Product structure and multiplicative extension problem. If a double complex K has a product structure $\cup$, relative to which its differential D is an antiderivation, the same is true of all the groups E_r and their operator d_r , since E_r is the homology of E_{r-1} and d_r is induced from D . With product structures, we have

³In fact, it is almost clear from the definition: You can regard an element in $H_\delta^p(\mathcal{U}, \mathcal{H}^q)$ as two parts, one eats an intersection of $(p+1)$ -fold, and the other outputs a q -form, that's how you get this isomorphism.

is a generator of $E_2^{2,0} = H^2(\mathbb{CP}^2) \otimes H^0(S^1)$, where x is a generator of $H^2(\mathbb{CP}^2)$. Then $x \otimes a$ is a generator of

$$E_2^{2,1} = H^2(\mathbb{CP}^2) \otimes H^1(S^1)$$

Thus a generator of $E_2^{4,0} = H^4(\mathbb{CP}^2)$ is given by

$$\begin{aligned} d_2(x \otimes a) &= d_2(x \otimes 1) \cdot (1 \otimes a) + (-1)^2(x \otimes 1) \cdot d_2(1 \otimes a) \\ &= x^2 \otimes 1 \end{aligned}$$

which implies x^2 is a generator of $H^4(\mathbb{CP}^2)$. So the ring structure of $\mathbb{CP}^2$ is

$$H^*(\mathbb{CP}^2) = \mathbb{R}[x]/(x^3)$$

where $|x| = 2$.

Remark 5.3. The same argument shows $H^*(\mathbb{CP}^n) = \mathbb{R}[x]/(x^{n+1})$, where $|x| = 2$.

In above example, we compute the cohomology ring of $\mathbb{CP}^2$ using fiber bundle $S^1 \rightarrow S^5 \rightarrow \mathbb{CP}^2$, in which $\mathbb{CP}^2$ is the base space. However, when we want to use spectral sequence to compute cohomology ring of total space of some fiber bundle, we need to be careful, since there is so-called multiplicative extension problem. To be explicit, there are two product structures on E_∞ -page:

1. The one inherited from double complex K ;
2. The one induced by fibration as follows: For elements $\bar{a}, \bar{b}$ belonging to $E_\infty^{p,q} = F^p H^{p+q}/F^{p+1} H^{p+q}$ and $E_\infty^{p',q'} = F^{p'} H^{p'+q'}/F^{p'+1} H^{p'+q'}$ respectively, we set

$$\bar{a} \cup \bar{b} := \overline{a \cup b}$$

in $E_\infty^{p+p',q+q'}$. Here $a \cup b$ makes sense since there is a product structure on $H^*(E)$.

We want these two product structures to be the same, since the first one is computable, and the second one is the product structure we want. However, something bad may happen, such as the following example:

Example 5.6. Consider the fiber bundle $S^2 \rightarrow \mathbb{CP}^3 \rightarrow S^4$, the E_2 -page of Leray spectral sequence is

2	$\mathbb{R}$				$\mathbb{R}$	
1						
0	$\mathbb{R}$				$\mathbb{R}$	
		0	1	2	3	4

By dimension reasons it is clear that this spectral sequence E_2 degenerates, that is $E_2 = \cdots = E_\infty$. If we use x to denote the generator of $H^2(S^2)$ and use y to denote the generator of $H^4(S^4)$, it is clear the cohomology ring structure

of E_∞ -page is $\mathbb{R}[x, y]/(x^2, y^2)$, where $|x| = 2, |y| = 4$, which is not isomorphic to the cohomology ring structure of $\mathbb{CP}^3$.

Remark 5.4. One can also refer to Example 1.17 in page 29 of [Hat04] for another example.

So the key point is that when the product structure on E_∞ recovers the one on $H^*(E)$. The next theorem gives some special condition under which the product structure of E_∞ recovers the one on $H^*(E)$.

Theorem 5.2. Suppose the associated graded algebra $E_\infty \cong \text{gr}_F H^*(E)$ is a free graded-commutative algebra on homogeneous classes, and each generator admits a lift to $H^*(E)$. Then these lifts identify the corresponding free algebra with $H^*(E)$; in particular, there are no multiplicative extension problems.

Proof. See Example in Page 25 of [McC00]. $\square$

5.3. Gysin sequence. In special cases the spectral sequence simplifies to a long exact sequence. One special case is the cohomology of sphere bundle and the resulting sequence is called Gysin sequence.

Let $\pi: E \rightarrow M$ be an oriented sphere bundle with fiber S^k . By assumption of orientability, there is no monodromy of locally constant sheaf $\mathcal{H}^k$, thus the E_2 -page of Leray spectral sequence is $H^p(M) \otimes H^q(S^k)$. Note that for arbitrary integer $n \geq k$, nothing in $E_2^{n-k, k}$ can be killed, thus there is an exact sequence

$$0 \rightarrow E_\infty^{n-k, k} \rightarrow E_2^{n-k, k}$$

and it can be extended to the following exact sequence

$$0 \rightarrow E_\infty^{n-k, k} \rightarrow E_2^{n-k, k} \xrightarrow{d_{k+1}} E_2^{n+1, 0} \rightarrow E_\infty^{n+1, 0} \rightarrow 0$$

since d_{k+1} is the only possible non-trivial map. On the other hand, the filtration on $H^n(E)$ becomes

$$\underbrace{H^n(E) \supset E_\infty^{n, 0}}_{E_\infty^{n-k, k}} \supset 0$$

which gives another exact sequence

$$0 \rightarrow E_\infty^{n, 0} \rightarrow H^n(E) \rightarrow E_\infty^{n-k, k} \rightarrow 0$$

Putting these two exact sequences together, one obtains

$$\dots \rightarrow H^n(E) \rightarrow H^{n-k}(M) \rightarrow H^{n+1}(M) \rightarrow H^{n+1}(E) \rightarrow \dots$$

To be explicit, you can find the map $H^{n-k}(M) \rightarrow H^{n+1}(M)$ is given by wedging with the Euler class of E .

5.4. Other coefficients. Since the de Rham cohomology is a cohomology theory with real coefficients, it is also necessary to take torsion phenomena into account. In this section we give a quick review of singular (co)homology, and show that the preceding results on spectral sequences carry over to integer coefficients.

5.4.1. *Review of singular (co)homology.* From now on, we usually use X to denote a topological space.

Definition 5.1 (singular q -simplex). A singular q -simplex in X is a continuous map $s: \Delta_q \rightarrow X$, where Δ_q is standard q -simplex.

Definition 5.2 (singular q -chain with $\mathbb{Z}$ coefficients). A singular q -chain in X is a finite linear combination with integer coefficients of singular q -simplices.

Notation 5.1. All singular q -chains form an abelian group, denoted by $S_q(X; \mathbb{Z})$.

Definition 5.3 (boundary map). The boundary map ∂ is defined as follows

$$\begin{aligned} \partial_q: S_q(X; \mathbb{Z}) &\rightarrow S_{q-1}(X; \mathbb{Z}) \\ \sigma &\mapsto \sum_i (-1)^i \sigma| [v_0, \dots, \widehat{v_i}, \dots, v_q] \end{aligned}$$

where we identify $[v_0, \dots, \widehat{v_i}, \dots, v_q]$ with Δ^{q-1} .

Definition 5.4 (singular homology group $\mathbb{Z}$ coefficients). The q -th singular homology group $H_q(X; \mathbb{Z})$ is defined as

$$H_q(X; \mathbb{Z}) := \ker \partial_q / \text{im } \partial_{q+1}$$

Lemma 5.1 (Poincaré lemma). $H_q(\mathbb{R}^n; \mathbb{Z}) = 0$ for all $q > 0$.

Definition 5.5 (singular q -cochain with $\mathbb{Z}$ coefficients). The group of singular q -cochains is defined as

$$S^q(X; \mathbb{Z}) := \text{Hom}(S_q(X; \mathbb{Z}), \mathbb{Z})$$

with coboundary map d_q defined by

$$(d_q \omega)(c) = \omega(\partial_{q+1} c)$$

where $\omega \in S^q(X), c \in S_q(X)$.

Definition 5.6 (singular cohomology group with $\mathbb{Z}$ coefficients). The q -th singular cohomology group $H^q(X; \mathbb{Z})$ is defined as

$$H^q(X; \mathbb{Z}) := \ker d_q / \text{im } d_{q-1}$$

Remark 5.5. Replacing $\mathbb{Z}$ with any arbitrary abelian group G , you can define singular (co)homology group with coefficients G .

Proposition 5.1. Given an open covering of X , the following sequence is exact

$$0 \leftarrow S_q^{\mathfrak{U}}(X; G) \leftarrow \bigoplus_{\alpha_0} S_q(U_{\alpha_0}; G) \leftarrow \bigoplus_{\alpha_0 < \alpha_1} S_q(U_{\alpha_0 \alpha_1}; G) \leftarrow \dots$$

where G is an arbitrary abelian group and $S_q^{\mathfrak{U}}(X, G)$ is the group of $\mathfrak{U}$ -small singular q -chain. Furthermore, there is a chain homotopy between $S_q(X; G)$ and $S_q^{\mathfrak{U}}(X; G)$.

Corollary 5.1. Given an open covering of X , the following sequence is exact

$$0 \rightarrow S_{\mathfrak{U}}^q(X; G) \rightarrow \bigoplus_{\alpha_0} S^q(U_{\alpha_0}; G) \rightarrow \bigoplus_{\alpha_0 < \alpha_1} S^q(U_{\alpha_0 \alpha_1}; G) \rightarrow \dots$$

where G is an arbitrary abelian group and $S_q^{\mathfrak{U}}(X, G)$ is the group of $\mathfrak{U}$ -small singular q -chain.

Theorem 5.3 (de Rham theorem). The singular cohomology with coefficients $\mathbb{R}$ is isomorphic to de Rham cohomology on smooth manifold.

Proof. Consider the double complex $C^*(\mathfrak{U}, S^*(\mathfrak{U}; \mathbb{R}))$, we can show ech cohomology of constant sheaf $\mathbb{R}$ is isomorphic to singular cohomology with coefficients $\mathbb{R}$, and we also know ech cohomology of constant sheaf $\mathbb{R}$ is isomorphic to de Rham cohomology. $\square$

Remark 5.6. In fact, for a topological space X whose good covers are cofinal, we can show ech cohomology of constant sheaf G is isomorphic to singular cohomology with coefficients G .

Notation 5.2. From now on, unless otherwise specified, we use $H^*(X)$ to denote cohomology with $\mathbb{Z}$ coefficients, and we use $H^*(X; -)$ to specify. For example, $H^*(X; \mathbb{R})$ denotes the $\mathbb{R}$ coefficients singular cohomology, that's also de Rham cohomology.

Theorem 5.4 (Leray spectral sequence for singular cohomology with coefficients in a commutative ring A). Let $\pi: E \rightarrow X$ be a fiber bundle with fiber F over a topological space X and $\mathfrak{U}$ an open covering of X . There is a spectral sequence converging to $H^*(E; A)$ with E_2 -term

$$E_2^{p,q} = H^p(\mathfrak{U}, \mathcal{H}^q(F; A))$$

Each E_r in the spectral sequence can be given a product structure relative to which the differential d_r is an antiderivation. If X is simply-connected and has a good cover, then

$$E_2^{p,q} = H^p(X, H^q(F; A))$$

Furthermore, if $H^*(F; A)$ is a finitely generated free A -module, then

$$E_2 = H^*(X; A) \otimes H^*(F; A)$$

as algebras over A .

Remark 5.7. Of course there is Leray spectral sequence for singular homology with coefficients, one simply reverses the arrows in the above case, here we omit the statement of it.

6. COHOMOLOGY OF SOME LIE GROUPS

A basic fact in differential geometry is that if G is a Lie group and H is a closed subgroup of G , then there exists the following fibration

$$\begin{array}{ccc} H & \longrightarrow & G \\ & & \downarrow \\ & & G/H \end{array}$$

If we are familiar with G/H and H , then the above fibration is a good way to compute cohomology ring of G . In fact, we always use the view of group action to give an explicit description of G/H .

6.1. Cohomology rings of $U(n)$ and $SU(n)$. Note that $U(n)$ acts on S^{2n-1} with stabilizer $U(n-1)$, that is $U(n)/U(n-1) = S^{2n-1}$, thus we have the following fibration:

$$\begin{array}{ccc} U(n-1) & \longrightarrow & U(n) \\ & & \downarrow \\ & & S^{2n-1} \end{array}$$

The same fibration still holds if we replace $U(n)$ by $SU(n)$.

Proposition 6.1. The cohomology ring of $U(n)$ is $\Lambda[x_1, \dots, x_{2n-1}]$, where $|x_i| = i, 1 \leq i \leq 2n-1$.

Proof. Note that $U(1) = S^1$, thus cohomology ring of $U(1)$ is $\Lambda[x_1]$, where $|x_1| = 1$. Apply Leray spectral sequence fibration

$$\begin{array}{ccc} U(n-1) & \longrightarrow & U(n) \\ & & \downarrow \\ & & S^{2n-1} \end{array}$$

we have E_2 -page has only two columns, that is $p = 0$ and $p = 2n-1$. Furthermore by induction we have cohomology ring of $U(n-1)$ is $\Lambda[x_1, \dots, x_{2n-3}]$, where $|x_i| = i, 1 \leq i \leq 2n-3$. Although there may be many nonzero rows of E_2 -page, but it suffices to check d_2 on those generators, that is the ones on $p = 0, q = 0, 1, 3, \dots, 2n-3$.

By dimension reasons, it is clear this spectral sequence degenerates at E_2 -page, which implies cohomology group structure of $U(n)$ is clear. If we choose a generator of $E_2^{2n-1,0}$, denoted by x_{2n-1} , then we can write the generator of $E_2^{2n-1,i}$ through product $E_2^{0,i} \times E_2^{2n-1,0} \rightarrow E_2^{2n-1,i}$. This shows that the cohomology ring of $U(n)$ is exactly $\Lambda[x_1, \dots, x_{2n-1}]$. $\square$

Remark 6.1. Note that here we can compute the cohomology ring structure of total space $U(n)$, since $\Lambda[x_1, \dots, x_{2n-1}]$ is free as an algebra.

Proposition 6.2. The cohomology ring of $SU(n)$ is $\Lambda[x_3, \dots, x_{2n-1}]$, where $n \geq 2, |x_i| = i, 1 \leq i \leq 2n-1$.

Proof. Note that $SU(2) = S^3$, thus cohomology ring of $SU(2)$ is $\Lambda[x_3]$, where $|x_3| = 3$. Apply Leray spectral sequence fibration

$$\begin{array}{ccc} SU(n-1) & \longrightarrow & SU(n) \\ & & \downarrow \\ & & S^{2n-1} \end{array}$$

The same argument shows the desired result. $\square$

6.2. Cohomology group of $SO(4)$. In this section we need the following fact.

Proposition 6.2.1. For a compact orientable manifold M , the integral $\int_M e(TM)$ is equal to its Euler number, that is $\sum (-1)^q H^q(M)$.

Example 6.1 (The cohomology ring of the unit tangent bundle of a sphere). The unit tangent bundle $S(T_{S^2})$ to the S^2 is a fiber bundle with fiber S^1 , that is

$$\begin{array}{ccc} S^1 & \longrightarrow & S(T_{S^{n-1}}) \\ & & \downarrow \\ & & S^2 \end{array}$$

If we consider $\mathbb{Z}_2$ coefficients, then the E_2 -page of the Leray spectral sequence is $E_2^{p,q} = H^p(S^2) \otimes H^q(S^1)$, that is

$$\begin{array}{c|ccc} & & \mathbb{Z}_2 & \mathbb{Z}_2 \\ 1 & & \searrow & \\ & \mathbb{Z}_2 & & \mathbb{Z}_2 \\ 0 & & & \\ \hline & 0 & 1 & 2 \end{array}$$

In order to compute E_3 , it suffices to compute above $d_2: E_2^{0,1} \rightarrow E_2^{2,0}$, and we know it defines the Euler class of $S(T_{S^2})$. By Proposition 6.2.1, the Euler class of $S(T_{S^2})$ is twice the generator of $H^2(S^2)$, then d_2 is zero, which implies this spectral sequence E_2 degenerates. Thus the cohomology group

$$H^q(S(T_{S^2})) = \begin{cases} \mathbb{Z}_2, & q = 0, 1, 2, 3 \\ 0, & \text{otherwise} \end{cases}$$

Remark 6.2. Here are two ways to think about $S(T_{S^2})$:

1. A point in $S(T_{S^2})$ is specified by a unit vector in $\mathbb{R}^3$ and another unit vector orthogonal to it, which can be completed to a unique orthonormal basis with positive determinant. Therefore $S(T_{S^2}) \cong SO(3)$
2. $SO(3)$ is the group of all rotations about the origin in $\mathbb{R}^3$, and each rotation is determined by its axis and an angle $-\pi \leq \theta \leq \pi$. In this way $SO(3)$ is parametrized by the solid 3-ball D^3 of radius π in $\mathbb{R}^3$. Furthermore,

antipodal points are glued together, since rotating through the angle $-\pi$ is the same as through π . Therefore $S(T_{S^2}) \cong \text{SO}(3)$.

Remark 6.3. Furthermore, since we can regard $S(T_{S^2})$ as $\mathbb{RP}^3$, it is clear that the above spectral sequence has multiplicative extension problem, since the ring structure of E_∞ -page is $\Lambda_{\mathbb{Z}_2}[x_1, x_2]$, where $|x_1| = 1, |x_2| = 2$ (The reason is the same as Example 5.6). However, the cohomology ring $H^*(\mathbb{RP}^3, \mathbb{Z}_2) = \mathbb{Z}_2[x]/(x^4)$, where $|x| = 1$.

Example 6.2 (The cohomology group of $\text{SO}(4)$). The group $\text{SO}(n)$ acts on S^{n-1} transitively with stabilizer $\text{SO}(n-1)$, therefore $\text{SO}(n)/\text{SO}(n-1) = S^{n-1}$. Thus we can use the Leray spectral sequence for the fiber bundle $\text{SO}(3) \rightarrow \text{SO}(4) \rightarrow S^3$. The E_2 -page is

3	$\mathbb{Z}$		$\mathbb{Z}$		
2	$\mathbb{Z}_2$		$\mathbb{Z}_2$		
1					
0	$\mathbb{Z}$		$\mathbb{Z}$		
		0	1	2	3

It is easy to see $d_2 = d_3 = \dots = 0$, which implies the cohomology group of $\text{SO}(4)$ is

$$H^q(\text{SO}(4)) = \begin{cases} \mathbb{Z} & q = 0, 6 \\ \mathbb{Z}_2 & q = 2, 5 \\ \mathbb{Z} \oplus \mathbb{Z} & q = 3 \\ 0 & \text{otherwise} \end{cases}$$

since there is no additive extension problem.

Example 6.3. Consider a manifold

$$W := \text{SU}(3)/\text{SO}(3)$$

where $\text{SO}(3)$ is embedded as a closed subgroup of $\text{SU}(3)$. Now we are going to compute its $\mathbb{Z}_2$ coefficients cohomology ring. It is clear that W is simply-connected, thus by Poincaré duality one has

$$H^1(W) = H^4(W) = 0, \quad H^2(W) = H^3(W)$$

Note that there is a fiber bundle $\text{SO}(3) \rightarrow \text{SU}(3) \rightarrow W$. Since $\text{SO}(3)$ is diffeomorphic to $\mathbb{RP}^3$, with the cohomology ring $\mathbb{Z}_2[x]/(x^4)$ and the cohomology ring of $\text{SU}(3)$ is $\Lambda[x_3, x_5]$, where $|x_i| = i$, the E_2 -page looks like

3	$\mathbb{Z}_2$		A	A		B	
2	$\mathbb{Z}_2$		A	A		B	
1	$\mathbb{Z}_2$		A	A		B	
0	$\mathbb{Z}_2$		A	A		B	
		0	1	2	3	4	5

where $A = H^2(W) = H^3(W)$, $B = H^5(W)$. In E_2 -page, we have the following observations:

1. $d_2: E_2^{0,1} \rightarrow E_2^{2,0}$ is isomorphism. Indeed, if this d_2 is not injective, then its kernel would survive to the E_∞ -page, that is $E_\infty^{0,1} \neq 0$, a contradiction to $H^1(\text{SU}(3)) = 0$. The same argument shows cokernel of it is also trivial.
2. $d_2: E_2^{3,1} \rightarrow E_2^{5,0}$ is an isomorphism by the same argument as above.

Now let's consider the cohomology ring structure of W .

1. Since $d_2: E_2^{0,1} \rightarrow E_2^{2,0}$ is an isomorphism, then we can use $x_2 \otimes 1$ to denote $d_2(1 \otimes a)$, then x_2 is a generator of $H^2(W)$;
2. Let x_3 be a generator of $H^3(W)$, since $d_2: E_2^{3,1} \rightarrow E_2^{5,0}$ is an isomorphism, and $x_3 \otimes a$ is a generator of $E_2^{3,1}$, then

$$\begin{aligned}
 d_2(x_3 \otimes a) &= d_2(x_3 \otimes 1) \cdot (1 \otimes a) + (-1)^3(x_3 \otimes 1) \cdot d_2(1 \otimes a) \\
 &= -(x_3 x_2 \otimes 1) \\
 &= (x_2 x_3 \otimes 1)
 \end{aligned}$$

which implies $x_2 x_3$ is a generator of $H^5(W)$.

Therefore, the cohomology ring of W is $\Lambda[x_2, x_3]$.

7. PATH FIBRATION

Recall that for a fiber bundle (E, X, F) , where E, X, F are topological spaces and X admits a good cover, then the E_2 -page of Leray's spectral sequence is

$$E_2^{p,q} = H^p(\mathcal{U}, \mathcal{H}^q(F))$$

where $\mathcal{H}^q(F)$ is a locally constant sheaf. Now suppose $\pi: E \rightarrow X$ is just a map, not necessarily locally trivial, we can still obtain a spectral sequence with E_2 -page $H^p(\mathcal{U}, \mathcal{H}^q(F))$ which converges to $H_D(E)$ as long as $\pi: E \rightarrow X$ has the property that $H^q(\pi^{-1}U) \cong H^q(F)$ for some fixed F and for all contractible open subset U , which is called path fibration property. An important example is path fibration.

7.1. Basic setting. Let X be a topological space with a base point $*$ and $[0, 1]$ the unit interval with base point 0. The path space of X is defined to be the space $P(X)$ consisting of all the paths in X with initial point $*$, that is

$$P(X) := \{\text{maps } \mu: [0, 1] \rightarrow X \mid \mu(0) = *\}$$

The path space $P(X)$ is equipped with compact-open topology; that is, a basis consists of all base-point preserving maps $\mu: [0, 1] \rightarrow X$ such that $\mu(K) \subset U$ for a fixed compact set K in $[0, 1]$ and a fixed open set U in X .

There is a natural projection $\pi: P(X) \rightarrow X$, defined by $\pi(\mu) = \mu(1)$. Now we claim $\pi: P(X) \rightarrow X$ has path fibration property. Indeed, for arbitrary contractible open set U containing p , there is a natural inclusion

$$i: \pi^{-1}(p) \rightarrow \pi^{-1}(U)$$

Since U is contractible, we can get a map

$$\phi: \pi^{-1}(U) \rightarrow \pi^{-1}(p)$$

It is clear $i \circ \phi = \text{id}$, and $\phi \circ i$ is homotopic to id , which implies $\pi^{-1}(U)$ has the same homotopy type as $\pi^{-1}(p)$. Furthermore, if p and q are in the same path component of X , then a fixed path from p to q gives a homotopy equivalence $\pi^{-1}(p) \cong \pi^{-1}(q)$. Thus all fibers have the homotopy type of $\pi^{-1}(*)$, which is loop space ΩX of X . To be explicit,

$$\Omega X = \{\mu: [0, 1] \rightarrow X \mid \mu(0) = \mu(1) = *\}$$

Thus $\pi: P(X) \rightarrow X$ has the path fibration property, that is $H^q(\pi^{-1}U) \cong H^*(\Omega X)$. Furthermore, path space PX is always contractible, since we can always shrink every path to its initial point.

Proposition 7.1. Let $\pi: E \rightarrow X$ be a path fibration. If X is simply-connected and E is path-connected, then the fibers are path-connected.

Proof. Trivially the $E_2^{0,0}$ term survives to E_∞ , hence

$$E_2^{0,0} = E_\infty^{0,0} = H^0(E) = \mathbb{Z}$$

since E is path-connected. On the other hand,

$$E_2^{0,0} = H^0(X, H^0(F)) = H^0(F)$$

which implies F is path-connected. $\square$

In fact there is a more general class of maps satisfying path fibration property, which is called a fibration. To be explicit, a map $\pi: E \rightarrow X$ is called a fibration if it satisfies the following property:

Theorem 7.1 (covering homotopy property). Given a map $f: Y \rightarrow E$ from any topological space Y into E and a homotopy $\bar{f}_t$ of $\bar{f} = \pi \circ f$, there is a homotopy f_t of f such that $\pi \circ f_t = \bar{f}_t$.

$$\begin{array}{ccc} Y & \xrightarrow{f} & E \\ \downarrow & \nearrow f_t & \downarrow \pi \\ Y \times I & \xrightarrow{\bar{f}_t} & X \end{array}$$

Proposition 7.2.

1. Any two fibers of a fibration over an arcwise-connected space have the same homotopy type;
2. For every contractible open set U , the inverse image $\pi^{-1}U$ has the homotopy type of the fiber F_a , where a is any point in U .

Proof. Here we only explain some key ideas in the proof of (1), which will be used later.

1. A path γ from a to b may be regarded as a homotopy of the point a ;
2. Let $\bar{g}: F_a \times I \rightarrow X$ be given by $(y, t) \mapsto \gamma(t)$, then covering homotopy property implies there exists a map $g: F_a \times I \rightarrow E$ that covers $\bar{g}$. Furthermore, $g_1 := g|_{F_a \times \{1\}}$ is a map from F_a to F_b , since $\gamma(1) = b$. Thus a path from a to b induces a map from F_a to the fiber F_b .
3. The **key point** is that homotopic paths from a to b in X induces homotopic maps from F_a to F_b .
4. If (3) holds, given $a, b \in X$ and a path γ from a to b , let $u: F_a \rightarrow F_b$ be a map induced by γ and $v: F_b \rightarrow F_a$ a map induced by γ^{-1} . Since $\gamma^{-1} \circ \gamma$ is homotopic to the constant map to a , the composition $v \circ u$ is homotopic to identity on F_a , which implies F_a and F_b have the same homotopy type.

$\square$

Remark 7.1. In fact, we can slightly change the proof to see if $\bar{f}_t, \bar{g}_t: Y \times I \rightarrow X$ are two homotopic homotopies, then their lifts $f_t, g_t: Y \times I \rightarrow E$ are also homotopic.

7.2. The cohomology ring of ΩS^n .

7.2.1. The cohomology group structure. In this section, we compute the integer cohomology groups of the loop space $\Omega S^n, n \geq 2$.

Example 7.1 (The cohomology group of ΩS^2). Since S^2 is simply-connected, the spectral sequence of the path fibration

$$\begin{array}{ccc} \Omega S^2 & \longrightarrow & PS^2 \\ & & \downarrow \\ & & S^2 \end{array}$$

has E_2 -page $H^p(S^2, H^q(\Omega S^2)) = H^p(S^2) \otimes H^q(\Omega S^2)$, thus only two nonzero columns at $p = 0, 2$. By dimensional reason, $d_3 = d_4 = \dots = 0$, thus $E_3 = E_\infty$. Furthermore, since PS^2 is contractible, we have all nonzero d_2 are isomorphisms. Thus $d_2: E_2^{0,1} \rightarrow E_2^{2,0}$ is an isomorphism, that is $H^1(\Omega S^2) = \mathbb{Z}$, but then

$$E_2^{2,1} = H^2(S^2) \otimes H^1(\Omega S^2) = \mathbb{Z}$$

by the same reason $E_2^{0,2} = \mathbb{Z}$. Step by step we find $H^q(\Omega S^2) = \mathbb{Z}$ in every dimension q .

Example 7.2 (The cohomology group of ΩS^3). Since S^3 is simply-connected, the spectral sequence of the path fibration

$$\begin{array}{ccc} \Omega S^3 & \longrightarrow & PS^3 \\ & & \downarrow \\ & & S^3 \end{array}$$

has E_2 -page $H^p(S^3, H^q(\Omega S^3)) = H^p(S^3) \otimes H^q(\Omega S^3)$, thus only two nonzero columns at $p = 0, 3$. By dimensional reason, $d_2 = d_4 = \dots = 0$, thus $E_3 = E_\infty$. Furthermore, since PS^3 is contractible, we have all nonzero d_3 are isomorphisms. Thus $d_3: E_2^{0,2} \rightarrow E_2^{3,0}$ is an isomorphism, that is $H^2(\Omega S^3) = \mathbb{Z}$, but then

$$E_2^{3,2} = H^3(S^3) \otimes H^2(\Omega S^3) = \mathbb{Z}$$

by the same reason $E_2^{0,4} = \mathbb{Z}$. Step by step we find $H^q(\Omega S^3) = \mathbb{Z}$ in every even dimension q .

Example 7.3. In general

$$H^q(\Omega S^n) = \begin{cases} \mathbb{Z}, & q = n-1, 2(n-1), \dots \\ 0, & \text{otherwise} \end{cases}$$

7.2.2. The cohomology ring structure. In this section, we compute the cohomology rings of the loop space $\Omega S^n, n \geq 2$.

Example 7.4 (The cohomology ring of ΩS^2). Let u be a generator of $E_2^{2,0} = H^2(S^2)$ and x a generator of $H^1(\Omega S^2)$ such that $d_2(1 \otimes x) = u \otimes 1$, then $u \otimes x$ is a generator of $H^2(S^2) \otimes H^1(\Omega S^2)$. Direct computation shows

$$\begin{aligned} d_2(1 \otimes x^2) &= d_2(1 \otimes x) \cdot (1 \otimes x) - (1 \otimes x) \cdot d_2(1 \otimes x) \\ &= (u \otimes 1) \cdot (1 \otimes x) - (1 \otimes x) \cdot (u \otimes 1) \\ &= u \otimes x - u \otimes x \\ &= 0 \end{aligned}$$

which implies $x^2 = 0$, since d_2 is an isomorphism. Let e be a generator of $H^2(\Omega S^2)$ such that $d_2(1 \otimes e) = u \otimes x$ and $u \otimes e \in H^2(S^2) \otimes H^2(\Omega S^2)$, then

$$\begin{aligned} d_2(1 \otimes ex) &= d_2(1 \otimes e) \cdot (1 \otimes x) + (1 \otimes e) \cdot d_2(1 \otimes x) \\ &= (u \otimes x) \cdot (1 \otimes x) + (1 \otimes e) \cdot (u \otimes 1) \\ &= u \otimes e \end{aligned}$$

implies ex is a generator of $H^3(\Omega S^2)$, since d_2 is an isomorphism. Similar computations show

$$\begin{aligned} d_2(1 \otimes \frac{e^2}{2}) &= \frac{1}{2} d_2(1 \otimes e) \cdot (1 \otimes e) + \frac{1}{2} (1 \otimes e) \cdot d_2(1 \otimes e) \\ &= \frac{1}{2} (u \otimes x) \cdot (1 \otimes e) + \frac{1}{2} (1 \otimes e) \cdot (u \otimes x) \\ &= (u \otimes ex) \\ d_2(1 \otimes \frac{e^2 x}{2}) &= \frac{1}{2} d_2(1 \otimes e^2) \cdot (1 \otimes x) + \frac{1}{2} (1 \otimes e^2) \cdot d_2(1 \otimes x) \\ &= (u \otimes ex) \cdot (1 \otimes x) + \frac{1}{2} (1 \otimes e^2) (u \otimes 1) \\ &= (u \otimes \frac{e^2}{2}) \end{aligned}$$

which implies $\frac{e^2}{2}$ is a generator of $H^4(\Omega S^2)$ and $\frac{e^2 x}{2}$ is a generator of $H^2(\Omega S^2)$. By induction we can show $\frac{e^k}{k!}$ is a generator of $H^{2k}(\Omega S^2)$ and $\frac{e^k x}{k!}$ is a generator of $H^{2k+1}(\Omega S^2)$.

The divided polynomial algebra $\mathbb{Z}_\gamma(e)$ with generator e is the $\mathbb{Z}$ -algebra with additive basis $\{1, e, e^2/2!, e^3/3!, \dots\}$, then

$$H^*(\Omega S^2) = \Lambda[x_1] \otimes \mathbb{Z}_\gamma(e)$$

where $|x_1| = 1, |e| = 2$.

Remark 7.2. By the same argument one can show that, when n is even,

$$H^*(\Omega S^n) = \Lambda[x_{n-1}] \otimes \mathbb{Z}_\gamma(e)$$

where $|x_{n-1}| = n-1, |e| = 2(n-1)$.

Example 7.5 (The cohomology ring of ΩS^3). Let u be a generator of $E_2^{3,0} = H^3(S^3)$ and e a generator of $H^2(\Omega S^3)$ such that $d_2(1 \otimes e) = u \otimes 1$, then $u \otimes e$ is a generator of $H^3(S^3) \otimes H^2(\Omega S^3)$. The same computation as in the above case shows $\frac{e^2}{2}$ is a generator of $H^2(\Omega S^3)$, and by induction one has $\frac{e^k}{k!}$ is a generator of $H^{2k}(\Omega S^3)$, which implies

$$H^*(\Omega S^3) = \mathbb{Z}_\gamma(e)$$

where $|e| = 2$.

Remark 7.3. By the same argument one can show that, when n is odd,

$$H^*(\Omega S^n) = \mathbb{Z}_\gamma(e)$$

where $|e| = n-1$.

Part 3. Applications to homotopy theory

8. REVIEW OF HOMOTOPY THEORY

8.1. First properties. Let X be a topological space with base point $*$.

Definition 8.1 (q -th homotopy group). For $q \geq 1$, the q -th homotopy group $\pi_q(X)$ of X is defined to be the homotopy classes of maps from the q -cube I^q to X which send the faces $\dot{I}^q$ of I^q to the base point of X .

Remark 8.1. Equivalently, $\pi_q(X), q \geq 1$ may be regarded as the homotopy classes of base-point preserving maps from S^q to X .

Remark 8.2. $\pi_0(X)$ is defined to be the set of all path components of X , and for a manifold the path components are the same as the connected components. Although $\pi_0(X)$ is in general not a group, if G is a Lie group then $\pi_0(G)$ is a group.

Proposition 8.1. Basic properties:

1. $\pi_q(X \times Y) = \pi_q(X) \times \pi_q(Y)$;
2. $\pi_q(X)$ is abelian if $q \geq 1$;
3. If $\tilde{X}$ is the universal covering of X , then $\pi_q(X) = \pi_q(\tilde{X})$ for $q \geq 2$.
4. $\pi_{q-1}(\Omega X) = \pi_q(X)$ for $q \geq 2$.

Proof. For (4). Elements of $\pi_2(X)$ are given by maps of I^2 to X , which can be viewed as a map from I to ΩX , therefore $\pi_2(X) = \pi_1(\Omega X)$. The general case is similar. $\square$

Example 8.1. The homotopy groups of S^1 are

$$\pi_q(S^1) = \begin{cases} \mathbb{Z}, & q = 1 \\ 0, & q > 1 \end{cases}$$

Theorem 8.1 (long exact sequence of homotopy). Let $\pi: E \rightarrow X$ be a base-point preserving fibration with fiber F , then there is an exact sequence of homotopy groups as follows

$$\cdots \rightarrow \pi_q(F) \xrightarrow{i_*} \pi_q(E) \xrightarrow{\pi_*} \pi_q(X) \xrightarrow{\partial} \pi_{q-1}(F) \rightarrow \cdots \rightarrow \pi_0(E) \rightarrow \pi_0(X) \rightarrow 0$$

Remark 8.3. Here we only give the descriptions of these homomorphisms, readers may refer to other standard textbooks for exactness.

The maps i_*, π_* are induced by the inclusion $i: F \rightarrow E$ and projection $\pi: E \rightarrow X$ respectively, where we regard F as the fiber over the base-point $*$ of B . To describe ∂ we use the covering homotopy property of fibration. A map $\alpha: I^q \rightarrow B$ representing an element of $\pi_q(X)$ can be regarded as a homotopy of $\alpha|_{I^{q-1}}$ in X . Note that $\alpha|_{I^{q-1}}: (t_1, \dots, t_{q-1}, 0) \rightarrow * \in X$, then we take constant map $*$: $I^{q-1} \rightarrow E$ from I^{q-1} to the base-point of F as the map that covers $\alpha|_{I^{q-1}}$. By the covering homotopy property, there is a homotopy $\bar{\alpha}: I^q \rightarrow E$ which covers α such that $\bar{\alpha}|_{I^{q-1}} = *$. Then $\partial[\alpha]$ is the homotopy class of the map $\bar{\alpha}: (t_1, \dots, t_{q-1}, 1) \rightarrow F$. And the well-definedness follows from Remark 7.1.

8.2. Hurewicz theorem.

Theorem 8.2 (Hurewicz theorem). Let X be a path-connected space, then $H_1(X)$ is the abelianization of $\pi_1(X)$.

Remark 8.4. So simply-connected space X must have $H_1(X) = 0$; The converse statement is not true, although it is quite difficult to give a simple example. For example, you can take an arbitrary perfect group⁴ G (For example, $G = A_5$), then the space $K(G, 1)$, which will be defined later is what you want. However, in general you do not know what it looks like.

Theorem 8.3 (Hurewicz theorem). Let X be a simply-connected path-connected CW complex. Then the first non-trivial homotopy group and homology group occur in the same dimension and are equal.

Proof. Let n denote the first dimension such that $\pi_n(X) \neq 0$, now let's prove by induction on n . First consider the case $n = 2$. The E_2 -page of homology spectral sequence of the path fibration is

$$\begin{array}{ccc|ccc}
 & & & & & & \\
 & & & & & & \\
 1 & & H_1(\Omega X) & & & & \\
 & & \swarrow & & & & \\
 0 & \mathbb{Z} & & 0 & & H_2(X) & \\
 \hline
 & 0 & 1 & 2 & & &
 \end{array}$$

Thus $\pi_2(X) = \pi_1(\Omega X) = H_1(\Omega X) = H_2(X)$.

Now let n be any positive integer ≥ 3 , then in this case ΩX has the following properties:

1. It is a CW complex⁵;
2. It is simply-connected, since $\pi_1(\Omega X) = \pi_2(X) = 0$;
3. The dimension of the first non-trivial homotopy group of ΩX is $n - 1$, since $\pi_{q-1}(\Omega X) = \pi_q(X)$, $q \geq 2$.

Then we can apply induction hypothesis to ΩX , one has

$$H_q(\Omega X) = \begin{cases} 0, & q < n - 1 \\ \pi_{n-1}(\Omega X) = \pi_n(X), & q = n - 1 \end{cases}$$

On the other hand, the E_2 -page still implies $H_{q-1}(\Omega X) = H_q(X)$ for $2 \leq q \leq n$. Then

$$H_q(X) = \begin{cases} 0, & 1 \leq q < n \\ H_{n-1}(\Omega X) = \pi_n(X), & q = n \end{cases}$$

□

⁴A group G such that its abelianization is trivial is called perfect group.

⁵Not a trivial fact, it is a theorem proved by Milnor: The loop space of a CW complex is still a CW complex.

Remark 8.5. For the ech model above, the base should admit a good cover. Every CW complex has the homotopy type of a simplicial complex, and open stars after subdivision give a good cover of that model. Alternatively, one may use the singular-cochain Serre spectral sequence directly.

Example 8.2. It follows from Hurewicz theorem that

$$\pi_q(S^n) = \begin{cases} 0, & q < n \\ \mathbb{Z}, & q = n \end{cases}$$

8.3. the Bott periodicity theorem.

Example 8.3 (stable homotopy groups of $U(n)$). Consider the following fibration

$$\begin{array}{ccc} U(n-1) & \longrightarrow & U(n) \\ & & \downarrow \\ & & S^{2n-1} \end{array}$$

Then homotopy exact sequence implies

$$\cdots \rightarrow \pi_{q+1}(S^{2n-1}) \rightarrow \pi_q(U(n-1)) \rightarrow \pi_q(U(n)) \rightarrow \pi_q(S^{2n-1}) \rightarrow \cdots$$

Then for $q < 2n - 2$, one has

$$\pi_q(U(n-1)) = \pi_q(U(n))$$

these mutually isomorphic groups are called q -th **stable homotopy groups** of the unitary group. They're denoted briefly by $\pi_q(U)$.

Remark 8.6. However, how to compute these stable homotopy groups? Bott has the following theorem:

Theorem 8.4 (the Bott periodicity theorem). For $q \geq 1$,

$$\pi_{q-1}(U) \cong \pi_{q+1}(U)$$

From this theorem, it suffices to compute $\pi_0(U)$ and $\pi_1(U)$, and it is quite clear:

$$\pi_0(U) = \pi_0(U(1)) = 0$$

$$\pi_1(U) = \pi_1(U(1)) = \mathbb{Z}$$

Example 8.4 (stable homotopy groups of $SU(n)$). For the same reason we have for $q < 2n$,

$$\pi_q(SU(n-1)) = \pi_q(SU(n))$$

and we also have q -th stable homotopy groups of the special unitary group, denoted by $\pi_q(SU)$. From the following fibration

$$\begin{array}{ccc} SU(n) & \longrightarrow & U(n) \\ & & \downarrow \det \\ & & S^1 \end{array}$$

we can conclude

$$\pi_q(\mathrm{U}(n)) = \begin{cases} \pi_q(\mathrm{SU}(n)), & q \geq 2 \\ \pi_1(\mathrm{SU}(n)) \oplus \mathbb{Z}, & q = 1 \end{cases}$$

for arbitrary $n \geq 1$. In particular, we have the isomorphisms between stable homotopy groups.

Example 8.5 (stable homotopy groups of $\mathrm{O}(n)$). Consider the following fibration

$$\begin{array}{ccc} \mathrm{O}(n-1) & \longrightarrow & \mathrm{O}(n) \\ & & \downarrow \\ & & S^{n-1} \end{array}$$

Then homotopy exact sequence implies

$$\cdots \rightarrow \pi_{q+1}(S^{n-1}) \rightarrow \pi_q(\mathrm{O}(n-1)) \rightarrow \pi_q(\mathrm{O}(n)) \rightarrow \pi_q(S^{n-1}) \rightarrow \cdots$$

Then for $q < n-2$, one has

$$\pi_q(\mathrm{O}(n-1)) = \pi_q(\mathrm{O}(n))$$

and we can define q -th stable homotopy groups of orthogonal groups, defined by $\pi_q(\mathrm{O})$. Similarly we also have the following theorem:

Theorem 8.5 (the Bott periodicity theorem). For $q \geq 0$,

$$\pi_q(\mathrm{O}) \cong \pi_{q+8}(\mathrm{O})$$

8.4. Some homotopy groups of Stiefel manifolds.

Definition 8.2 (real Stiefel manifolds). The real Stiefel manifolds $V_k(\mathbb{R}^{n+k})$ is the set of all orthonormal k -frames in $\mathbb{R}^{n+k}$.

Example 8.6. $\mathrm{SO}(n) = V_{n-1}(\mathbb{R}^n)$.

Example 8.7. $S^n = V_1(\mathbb{R}^{n+1})$.

Lemma 8.1. For $1 \leq k \leq n$, $V_k(\mathbb{R}^{n+k})$ is $(n-1)$ -connected, and

$$\pi_n(V_2(\mathbb{R}^{n+2})) = \begin{cases} \mathbb{Z}, & n \text{ is odd} \\ \mathbb{Z}/2\mathbb{Z}, & n \text{ is even} \end{cases}$$

Proof. Apply homotopy exact sequence to the following fibration

$$\begin{array}{ccc} V_{k-1}(\mathbb{R}^{n+k-1}) & \longrightarrow & V_k(\mathbb{R}^{n+k}) \\ & & \downarrow \\ & & S^{n+k-1} \end{array}$$

Then if $q < n+k-2$, one has

$$\pi_q(V_k(\mathbb{R}^{n+k})) = \pi_q(V_{k-1}(\mathbb{R}^{n+k-1}))$$

In particular if $q < n$, one has

$$\pi_q(V_k(\mathbb{R}^{n+k})) = \pi_q(V_{k-1}(\mathbb{R}^{n+k-1})) = \cdots = \pi_q(V_1(\mathbb{R}^{n+1})) = \pi_q(S^n) = 0$$

□

8.5. Hopf invariant.

8.5.1. *History.* In general, it is tough to compute $\pi_q(S^n)$ for $n \geq 2, q > n$. So the first non-trivial case is $\pi_3(S^2)$. Consider the Hopf fibration

$$\begin{array}{ccc} S^1 & \longrightarrow & S^3 \\ & & \downarrow \\ & & \mathbb{CP}^1 = S^2 \end{array}$$

Then the exact sequence of homotopy groups implies

$$\cdots \rightarrow \pi_q(S^1) \rightarrow \pi_q(S^3) \rightarrow \pi_q(S^2) \rightarrow \pi_{q-1}(S^1) \rightarrow \cdots$$

Use the fact that $\pi_q(S^1) = 0, q > 1$ one has

$$\pi_q(S^3) = \pi_q(S^2)$$

for $q > 1$. In particular one has $\pi_3(S^2) = \mathbb{Z}$. Historically, $\pi_3(S^2)$ was first computed by Hopf in 1931 using a linking number argument which associates to each homotopy class of maps from S^3 to S^2 an integer now called the Hopf invariant. We first give an account of the Hopf invariant in the dual language of differential forms and then in terms of the linking number.

8.5.2. The differential forms definition.

Definition 8.3 (Hopf invariant). Let $f: S^{2n-1} \rightarrow S^n$ be a smooth map and let α be a generator of $H_{dR}^n(S^n)$, then the Hopf invariant of f is defined as

$$H(f) = \int_{S^{2n-1}} \omega \wedge d\omega$$

where $f^* \alpha = d\omega$.

Proposition 8.2. Properties of the Hopf invariant:

1. The definition of Hopf invariant is independent of the choice of ω ;
2. For odd n the Hopf invariant is 0;
3. Homotopic maps have the same Hopf invariant.

Proof. For (1). Let ω' be another $(n-1)$ -form on S^{2n-1} such that $f^* \alpha = d\omega'$. Then

$$\begin{aligned} \int_{S^{2n-1}} \omega \wedge d\omega - \int_{S^{2n-1}} \omega' \wedge d\omega' &= \int_{S^{2n-1}} (\omega - \omega') \wedge d\omega \\ &= \pm \int_{S^{2n-1}} d((\omega - \omega') \wedge \omega) \\ &= 0 \end{aligned}$$

For (2). If n is odd, then ω is even-dimensional, thus

$$\omega \wedge d\omega = \frac{1}{2} d(\omega \wedge \omega)$$

For (3). From (2) we may assume n is even. Let $f: S^{2n-1} \times I \rightarrow S^n$ be a homotopy between $f_0, f_1: S^{2n-1} \rightarrow S^n$. We use i_0 to denote the inclusion $i_0: S^{2n-1} \rightarrow S_0 = S^{2n-1} \times \{0\} \subset S^{2n-1} \times I$ and similarly for i_1 . Then

$$F \circ i_0 = f_0$$

$$F \circ i_1 = f_1$$

Let α be a generator of $H_{dR}^n(S^n)$, then $F^* \alpha = d\omega$ for some $(n-1)$ -form ω on $S^{2n-1} \times I$. Define $i_0^* \omega = \omega_0$ and $i_1^* \omega = \omega_1$, then

$$f_0^* \alpha = (F \circ i_0)^* = i_0^* \circ F^* \alpha = \omega_0$$

$$f_1^* \alpha = (F \circ i_1)^* = i_1^* \circ F^* \alpha = \omega_1$$

Then

$$\begin{aligned} H(f_1) - H(f_2) &= \int_{S^{2n-1}} \omega_1 \wedge d\omega_1 - \int_{S^{2n-1}} \omega_0 \wedge d\omega_0 \\ &= \int_{S^{2n-1}} i_1^*(\omega \wedge d\omega) - \int_{S^{2n-1}} i_0^*(\omega \wedge d\omega) \\ &= \int_{S_1} - \int_{S_0} \omega \wedge d\omega \\ &= \int_{S^{2n-1}} d(\omega \wedge d\omega) \\ &= \int_{S^{2n-1} \times I} F^*(\alpha \wedge \alpha) \\ &= 0 \end{aligned}$$

□

Thus Hopf invariant gives the following map

$$H: \pi_{2n-1}(S^n) \rightarrow \mathbb{R}$$

Furthermore, it gives a group homomorphism. Indeed, for two smooth maps $f, g: S^{2n-1} \rightarrow S^n$, it suffices to show

$$(fg)^*(\alpha) =$$

where α is a generator of $H_{dR}^n(S^n)$ and $d\omega_f = f^* \alpha, d\omega_g = g^* \alpha$. Then

$$H(fg) = \int_{S^{2n-1}}$$

8.5.3. *The intersection-theory definition.*

9. APPLICATIONS TO HOMOTOPY THEORY

9.1. Eilenberg-MacLane spaces.

Definition 9.1 (Eilenberg-MacLane spaces). Let G be a group, a path-connected space X is called an Eilenberg-MacLane space $K(G, n)$, if

$$\pi_q(X) = \begin{cases} G, & q = n \\ 0, & \text{otherwise} \end{cases}$$

Remark 9.1. For any group G and $n \geq 1$, in the category of CW complexes $K(G, n)$ exists and is unique up to homotopy equivalence.

Example 9.1. S^1 is $K(\mathbb{Z}, 1)$ according to Example 8.1.

Example 9.2. If F is a free group, then $K(F, 1)$ is a connected graph.

Corollary 9.1. Any subgroup of a free group is still a free group.

Example 9.3. For a group G with generators $a_1, b_1, \dots, a_g, b_g$ and a single relation

$$a_1 b_1 a_1^{-1} b_1^{-1} \dots a_g b_g a_g^{-1} b_g^{-1} = 1$$

Then the Riemann surface with genus g is $K(G, 1)$, according to the following theorem.

Theorem 9.1 (uniformization theorem). Every simply-connected Riemann surface is biholomorphic to

1. S^2 ;
2. $\mathbb{C}$;
3. the unit disk Δ in $\mathbb{C}$.

For compact Riemann surfaces,

1. those with universal cover Δ are precisely the surfaces of genus greater than 1;
2. those with universal cover $\mathbb{C}$ are the Riemann surfaces of genus 1, namely the complex tori or elliptic curves;
3. those with universal cover S^2 are those of genus zero, namely the Riemann sphere itself.

Proposition 9.1. Basic properties:

1. $\Omega K(G, n) = K(G, n-1)$;
2. $K(G \times H, n) = K(G, n) \times K(H, n)$.

9.2. The telescoping construction. In this section we introduce a technique for constructing certain Eilenberg-MacLane spaces, which is called telescoping construction.

Example 9.4 (The infinite real projective space). Note that we have the following natural inclusions

$$\{\text{point}\} \hookrightarrow \dots \xhookrightarrow{i} \mathbb{R}P^n \xhookrightarrow{i} \mathbb{R}P^{n+1} \hookrightarrow \dots$$

Then we define the infinite real projective space $\mathbb{RP}^\infty$ as

$$\mathbb{RP}^\infty := \bigcup_n \mathbb{RP}^n$$

Since $S^n \rightarrow \mathbb{RP}^n$ is a double cover, $\pi_q(\mathbb{RP}^n) = 0$ for $1 < q < n$ and $\pi_1(\mathbb{RP}^n) = \mathbb{Z}_2$. We claim $\mathbb{RP}^\infty$ is exactly $K(\mathbb{Z}_2, 1)$.

1. For arbitrary $q > 1, f \in \pi_q(\mathbb{RP}^\infty)$, that is a map $f: S^q \rightarrow \mathbb{RP}^\infty$. Since S^q is compact, then there exists a sufficiently large N such that $f(S^q) \subset \mathbb{RP}^N$, and $\pi_q(\mathbb{RP}^N) = 0$ implies f is null-homotopic;
2. Similarly we can construct infinite sphere S^∞ , which is a double cover of $\mathbb{RP}^\infty$, and by the same argument we have S^∞ is contractible. Then homotopy exact sequence of fibration implies $\pi_1(\mathbb{RP}^\infty) = \mathbb{Z}_2$.

Example 9.5 (The infinite complex projective space). Note that we have the following natural inclusions

$$\{\text{point}\} \hookrightarrow \dots \hookrightarrow \mathbb{CP}^n \hookrightarrow \mathbb{CP}^{n+1} \hookrightarrow \dots$$

Then we define the infinite complex projective space $\mathbb{CP}^\infty$ as

$$\mathbb{CP}^\infty := \bigcup_n \mathbb{CP}^n$$

Similarly we have the following fibration

$$\begin{array}{ccc} S^1 & \longrightarrow & S^\infty \\ & & \downarrow \\ & & \mathbb{CP}^\infty \end{array}$$

By the same argument you can see $\mathbb{CP}^\infty = K(\mathbb{Z}, 2)$.

Proposition 9.2. The cohomology ring of $\mathbb{CP}^\infty$ is $\mathbb{Z}[x]$, where $|x| = 2$.

Proof. Note that $\mathbb{CP}^\infty$ is simply-connected, then the E_2 -page of Leray spectral sequence is

1	Z	0	A	B
0	Z	0	A	B
	0	1	2	3

and these d_2 are isomorphisms, which implies the cohomology group of $\mathbb{CP}^\infty$ is

$$H^q(\mathbb{CP}^\infty) = \begin{cases} \mathbb{Z}, & q \text{ is even} \\ 0, & \text{otherwise} \end{cases}$$

To see its ring structure, we rewrite E_2 -page as follows:

$$\begin{array}{c|ccccc}
1 & 1 \otimes u & & x \otimes u & & x^2 \otimes u \\
0 & 1 \otimes 1 & & x \otimes 1 & & x^2 \otimes 1 \\
\hline
& 0 & 1 & 2 & 3 & 4
\end{array}$$

$\swarrow \quad \quad \quad \searrow$

Take a generator u of $H^1(S^1)$ and use $x \otimes 1$ to denote $d_2(1 \otimes u)$, then x is a generator of $H^2(\mathbb{CP}^\infty)$. Then

$$\begin{aligned}
d_2(x \otimes u) &= d_2(x \otimes 1) \cdot (1 \otimes u) + (-1)^2(x \otimes 1) \cdot d_2(1 \otimes u) \\
&= (x^2 \otimes 1)
\end{aligned}$$

implies x^2 is a generator of $H^4(\mathbb{CP}^\infty)$. By induction you can see its cohomology ring is $\mathbb{Z}[x]$, where $|x| = 2$. $\square$

Example 9.6 (The infinite lens spaces). Since S^1 acts freely on S^{2m+1} , so does any subgroup of S^1 . Consider $\mathbb{Z}_n$ -action on S^{2m+1} as follows

$$\begin{aligned}
\mathbb{Z}_n \times S^{2m+1} &\rightarrow S^{2m+1} \\
(e^{\frac{2\pi i}{n}}, (z_0, \dots, z_m)) &\rightarrow (e^{\frac{2\pi i}{n}} z_0, \dots, e^{\frac{2\pi i}{n}} z_m)
\end{aligned}$$

The quotient space of S^{2m+1} by the action of $\mathbb{Z}_n$ is called lens space $L(m, n)$. Apply telescoping construction we can define infinite lens space $L(\infty, n)$, and there is a fibration

$$\begin{array}{ccc}
\mathbb{Z}_n & \longrightarrow & S^\infty \\
& & \downarrow \\
& & L(\infty, n)
\end{array}$$

By the same argument one can show $L(\infty, n)$ is $K(\mathbb{Z}_n, 1)$.

Remark 9.2. In particular, $L(\infty, 2)$ is exactly $\mathbb{RP}^\infty$.

In order to show the cohomology of $L(m, n)$, the fibration $\mathbb{Z}_n \rightarrow S^{2m+1} \rightarrow L(m, n)$ makes no sense, since $L(m, n)$ is not simply-connected. Instead, note that the free action of S^1 on S^{2m+1} descends to an action on $L(m, n)$:

$$(z_0, \dots, z_m) \mapsto (\lambda z_0, \dots, \lambda z_m)$$

since S^1 is an abelian Lie group. Furthermore, the quotient of this descended action is still $\mathbb{CP}^m$, so there is a fibration

$$\begin{array}{ccc}
S^1 & \longrightarrow & L(m, n) \\
& & \downarrow \\
& & \mathbb{CP}^m
\end{array}$$

and the E_2 -page of Leray spectral sequence to this fibration is

$$\begin{array}{c|ccccc}
1 & 1 \otimes u & & x \otimes u & & x^2 \otimes u \\
0 & 1 \otimes 1 & & x \otimes 1 & & x^2 \otimes 1 \\
\hline
& 0 & 1 & 2 & 3 & 4
\end{array}$$

$\swarrow \quad \searrow$
 $\quad \quad \quad \searrow$

Let u be a generator of $H^1(S^1)$ and x a generator of $H^2(\mathbb{CP}^m)$, it suffices to compute what does $d_2: E_2^{0,1} \rightarrow E_2^{2,0}$ look like, since $x^n \otimes a$ generates $E_2^{2n,1}$. However, since $\pi_1(L(m,n)) = \mathbb{Z}_n$, then Hurewicz theorem implies $H_1(L(m,n)) = \mathbb{Z}_n$, so universal coefficients theorem implies

$$H^1(L(m,n)) = \text{Hom}(H_1(L(m,n)), \mathbb{Z}) \oplus \text{Ext}(H_0(X), \mathbb{Z}) = 0$$

$$H^2(L(m,n)) = \text{Hom}(H_2(L(m,n)), \mathbb{Z}) \oplus \text{Ext}(H_1(X), \mathbb{Z}) = \text{free part} \oplus \mathbb{Z}_n$$

So we have d_2 is multiplication by n , and the cohomology groups of $L(m,n)$ are:

$$H^q(L(m,n)) = \begin{cases} \mathbb{Z}, & q = 0, 2m+1 \\ \mathbb{Z}_n, & q = 2, 4, \dots, 2m \\ 0, & \text{otherwise} \end{cases}$$

If we consider the following fibration

$$\begin{array}{ccc}
S^1 & \longrightarrow & K(\mathbb{Z}_n, 1) \\
& & \downarrow \\
& & \mathbb{CP}^\infty
\end{array}$$

Then the same computation shows the cohomology groups of $K(\mathbb{Z}_n, 1)$ are

$$H^q(K(\mathbb{Z}_n, 1)) = \begin{cases} \mathbb{Z}, & q = 0 \\ \mathbb{Z}_n, & q \text{ is even}, q \neq 0 \\ 0, & \text{otherwise} \end{cases}$$

However, if we consider $\mathbb{Q}$ coefficients, then multiplication by n is an isomorphism of $\mathbb{Q}$, which implies

$$H^q(K(\mathbb{Z}_n, 1); \mathbb{Q}) = \begin{cases} \mathbb{Q}, & k = 0 \\ 0, & \text{otherwise} \end{cases}$$

Furthermore, we can use path fibration of $K(\mathbb{Z}_n, 1)$ to conclude

$$H^q(K(\mathbb{Z}_n, m); \mathbb{Q}) = \begin{cases} \mathbb{Q}, & q = 0 \\ 0, & \text{otherwise} \end{cases}$$

holds for all $m \in \mathbb{Z}_{>0}$. Therefore, by the structure of finitely generated abelian groups, the rational cohomology of $K(G, m)$ is trivial for a finitely generated torsion abelian group.

9.3. Some results about cohomology ring of $K(\mathbb{Z}, 3)$. Since $\pi_q(S^3) = 0$ for $q < 3$ and $\pi_3(S^3) = \mathbb{Z}$, it is natural to ask whether S^3 is $K(\mathbb{Z}, 3)$ or not. Since we know $\Omega K(\mathbb{Z}, 3) = K(\mathbb{Z}, 2) = \mathbb{CP}^\infty$ and we have the following path fibration

$$\begin{array}{ccc} K(\mathbb{Z}, 2) & \longrightarrow & PK(\mathbb{Z}, 3) \\ & & \downarrow \\ & & K(\mathbb{Z}, 3) \end{array}$$

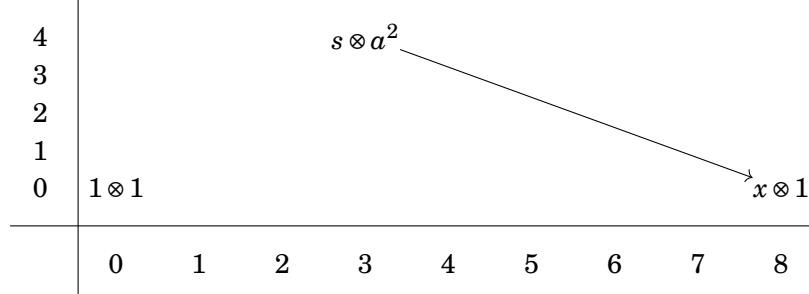
Then we can use Leray spectral sequence to compute cohomology ring of $K(\mathbb{Z}, 3)$. By dimensional reason, it is clear the E_2 -page equals the E_3 -page, which looks like

8	$1 \otimes a^4$															
7																
6	$1 \otimes a^3$															
5																
4	$1 \otimes a^2$															
3																
2	$1 \otimes a$															
1																
0	$1 \otimes 1$															
		$s \otimes 1$		$s \otimes a$		$s \otimes a^2$		$s \otimes a^3$		$s^2 \otimes a$		$s^2 \otimes a^2$		$s^2 \otimes a^3$		$s^3 \otimes 1$
		0	1	2	3	4	5	6	7	8	9					

where a is the generator of $H^2(\mathbb{CP}^\infty)$. We have the following observations:

1. It is clear $d_3: E_3^{0,2} \rightarrow E_3^{3,0}$ is an isomorphism, and we use $s \otimes 1$ to denote $d_3(1 \otimes a)$, then s is a generator of $H^3(K(\mathbb{Z}, 3))$.
2. $H^4(K(\mathbb{Z}, 3)) = 0$, otherwise a nonzero element in $E_3^{4,0}$ will live to E_∞ -page, since nothing can kill it; The same argument shows $H^5(K(\mathbb{Z}, 3)) = 0$.
3. Note that $s \otimes a$ is a generator of $E_3^{3,2}$, $s \otimes a^2$ is a generator of $E_3^{3,4}$ and $s \otimes a^3$ is a generator of $E_3^{3,6}$, then by antiderivation property of d_3 , we can see
 - (a) $d_3: E_3^{0,4} \rightarrow E_3^{3,2}$ is multiplication by 2;
 - (b) $d_3: E_3^{0,6} \rightarrow E_3^{3,4}$ is multiplication by 3;
 - (c) $d_3: E_3^{0,8} \rightarrow E_3^{3,6}$ is multiplication by 4.
4. $d_3: E_3^{3,2} \rightarrow E_3^{6,0}$ cannot be zero map, otherwise $E_\infty^{3,2} = \text{coker}\{d_3: E_3^{0,4} \rightarrow E_3^{3,2}\} = \mathbb{Z}_2$, a contradiction. Furthermore, it is also surjective, otherwise $E_\infty^{6,0} = \text{coker}\{d_3: E_3^{3,2} \rightarrow E_3^{6,0}\} \neq 0$, also a contradiction. Combine these two facts together one has $E_3^{6,0} = \mathbb{Z}_2$ is generated by $s^2 \otimes 1$, that is s^2 is a generator of $H^6(K(\mathbb{Z}, 3))$.
5. Since $2s^2 = 0$, then $d_3: E_3^{3,4} \rightarrow E_3^{6,2} = 0$. Thus $E_4^{3,4} = \mathbb{Z}_3$, and there are no maps into $E_4^{3,4}$. So the only possible element to kill it is elements in $E_5^{8,0}$,

which implies $H^8(K(\mathbb{Z}, 3)) = \mathbb{Z}_3$, we use x to denote its generator. This process is shown in the following E_5 -page.

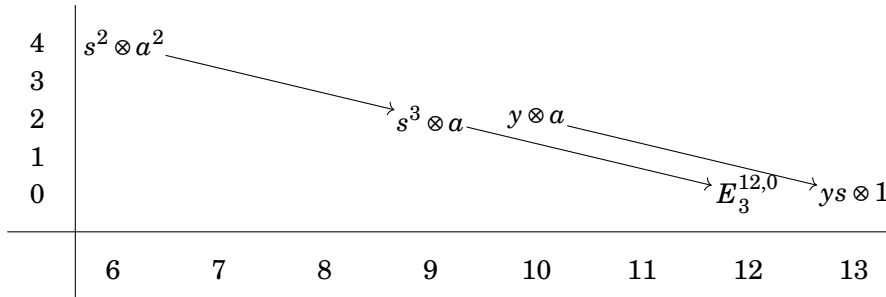


6. $d_3: E_3^{6,2} \rightarrow E_3^{9,0}$ is an isomorphism, and the reason is the same as (4), which implies s^3 is a generator of $H^9(K(\mathbb{Z}, 3))$.
7. Since $2s^2 = 0$, $d_3: E_3^{3,6} \rightarrow E_3^{6,4}$ is surjective, with kernel $2(s \otimes a^3)$, which implies $E_4^{3,6} = \mathbb{Z}_2$. Note that only the following two maps may kill this nonzero element.
 - (a) $d_5: E_5^{3,6} \rightarrow E_5^{8,2}$;
 - (b) $d_7: E_7^{3,6} \rightarrow E_7^{10,0}$.

Now let's consider case (a), this will happen if and only if $d_3: E_3^{8,2} \rightarrow E_3^{11,0}$ is a zero map, that is $E_5^{8,2} = \mathbb{Z}_3 \neq 0$. However, any group homomorphism from $\mathbb{Z}_2$ to $\mathbb{Z}_3$ must be trivial. So case (a) will not kill this $\mathbb{Z}_2$. Furthermore, it cannot happen, otherwise $E_\infty^{8,2} \neq 0$. So

- I We must have $d_7: E_7^{3,6} \rightarrow E_7^{10,0}$ is an isomorphism, which implies $H^{10}(K(\mathbb{Z}, 3)) = \mathbb{Z}_2$, and we use y to denote this generator;
- II By the way we conclude $d_3: E_3^{8,2} \rightarrow E_3^{11,0}$ is an isomorphism, which implies xs is a generator of $H^{11}(K(\mathbb{Z}, 3))$.

Now let's discard the information already determined and concentrate on something unknown to draw a new picture of E_3 -page as follows:



7. Since $2s^2 = 0$, $d_3: E_3^{6,4} \rightarrow E_3^{9,2}$ is zero map. So $d_3: E_3^{9,2} \rightarrow E_3^{12,0}$ is an injective, which implies there is a $s^4 \otimes 1 \in E_3^{12,0}$. However, $d_3: E_3^{9,2} \rightarrow E_3^{12,0}$ is not a surjective, since we also need $d_9: E_9^{3,8} \rightarrow E_9^{12,0}$ to kill $E_9^{3,8} = \mathbb{Z}_5$, which implies $H^{12}(K(\mathbb{Z}, 3)) = \mathbb{Z}_2 \oplus \mathbb{Z}^5$, one of the generators is s^4 , and the other one we use z to denote it.

8. Finally one can check $d_3: E_3^{10,2} \rightarrow E_3^{13,0}$ is an isomorphism, which implies $H^{13}(K(\mathbb{Z}, 3))$ is generated by ys .

In summary the first few cohomology groups of $K(\mathbb{Z}, 3)$ are

q	0	1	2	3	4	5	6	7	8	9	10	11	12	13
$H^q(K(\mathbb{Z}, 3))$	$\mathbb{Z}$	0	0	$\mathbb{Z}$	0	0	$\mathbb{Z}_2$	0	$\mathbb{Z}_3$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_2 \oplus \mathbb{Z}_5$	$\mathbb{Z}_2$
generators	1			s			s^2		x	s^3	y	xs	s^4, z	ys

The bad news is, that is you cannot figure out what is $H^{14}(K(\mathbb{Z}, 3))$ using the above method, we do need additional information. However, the situation can be vastly simplified by taking coefficients in $\mathbb{Q}$ rather than $\mathbb{Z}$. In this case we have

Proposition 9.3.

$$H^*(K(\mathbb{Z}, n); \mathbb{Q}) = \begin{cases} \mathbb{Q}[x], & n \text{ is even} \\ \Lambda_{\mathbb{Q}}[x], & n \text{ is odd} \end{cases}$$

where $|x| = n$.

Proof. Let's prove by induction on n via the following path fibration

$$\begin{array}{ccc} K(\mathbb{Z}, n-1) & \longrightarrow & PK(\mathbb{Z}, n) \\ & & \downarrow \\ & & K(\mathbb{Z}, n) \end{array}$$

1. For $n = 2$, we have already shown $H^*(K(\mathbb{Z}, 2); \mathbb{Q}) = \mathbb{Q}[x]$, where $|x| = 2$;
2. For $n = 3$, we just need to replace $\mathbb{Z}$ with $\mathbb{Q}$ in above computation, and note that multiplication by i is an isomorphism of $\mathbb{Q}$. Then one can argue inductively that $E_3^{p,0}$ must be zero for $p > 3$, otherwise the first nonzero one will live to E_{∞} -page, since it cannot be killed by any differential.

The induction process is then routine. $\square$

Remark 9.3. More generally, this holds also when $\mathbb{Q}$ is replaced by any nonzero subring of $\mathbb{Q}$. See Proposition 1.20 on page 31 of [Hat04].

9.4. Basic tricks of the trade. In homotopy theory, every map $f: A \rightarrow B$ from a space A to a path-connected space B may be viewed as

1. An inclusion;
2. A fibration.

9.4.1. Inclusion. Consider the mapping cylinder of f as follows:

$$M_f := (A \times I) \cup B / (a, 1) \sim f(a)$$

It is clear that M_f has the same homotopy type as B and A is included in M_f .

9.4.2. Fibration. Without loss of generality we may assume f is an inclusion. Define L to be the space of all paths in B with initial point in A . By shrinking every path to its initial point, we get a homotopy equivalence $L \simeq A$. On the other hand, by projecting every path to its endpoint, we get the following fibration

$$\begin{array}{ccc} \Omega_*^A & \longrightarrow & L \simeq A \\ & & \downarrow \\ & & B \end{array}$$

whose fiber is Ω_*^A , the space of all paths from a point $*$ in B to A . So up to homotopy equivalence, $f: A \rightarrow B$ is a fibration.

9.5. Postnikov approximation. Let X be a CW complex with homotopy groups $\pi_q(X) = \pi_q$. Although X has the same homotopy groups as the product space $\prod K(\pi_q, q)$, in general it will not have the same homotopy type. However, up to homotopy every CW complex can be thought of as a twisted product" of Eilenberg-MacLane spaces in the following sense.

Proposition 9.4 (Postnikov approximation). Every CW complex can be approximated by a twisted product of Eilenberg-MacLane spaces; More precisely, for each n , there is a sequence of fibrations $Y_q \rightarrow Y_{q-1}$ with the $K(\pi_q, q)$'s as fibers and commuting maps $X \rightarrow Y_q$

$$\begin{array}{ccccccc} K(\pi_1, 1) = Y_1 & \xleftarrow{K(\pi_2, 2)} & Y_2 & \xleftarrow{\quad} & \dots & \xleftarrow{\quad} & Y_n \\ & \searrow & \swarrow & & & & \uparrow \\ & & & & & & X \end{array}$$

such that the map $X \rightarrow Y_q$ induces an isomorphism of homotopy groups in dimensions $\leq q$. Such a sequence of fibrations is called a Postnikov tower of X .

Remark 9.4. First let's explain a procedure for killing the homotopy groups of X above a given dimension. For example, to construct $K(\pi_1, 1)$ we kill off the homotopy groups of X in dimensions ≥ 2 .

If $\alpha: S^2 \rightarrow X$ represents a non-trivial element in $\pi_2(X)$, we attach a 3-cell to X via α as follows:

$$X \cup_\alpha e^3 = X \amalg e^3 / x \sim \alpha(x), \quad x \in S^2$$

This procedure does not change the fundamental group of the space, since attaching a n -cell to X could kill an element of $\pi_{n-1}(X)$, but does not affect the homotopy of X in dimensions $\leq n-2$. So for each generator of $\pi_2(X)$ we attach a 3-cell to X as above. In this way we create a new space X_1 with the same fundamental group as X but with no π_2 . Iterating this procedure we can kill all higher homotopy groups. This gives Y_1 .

Proof. To construct Y_n we kill off all homotopy groups of X in dimensions $\geq n+1$ by attaching cells. Then

$$\pi_q(Y_n) = \begin{cases} 0, & q \geq n+1 \\ \pi_q, & q = 1, 2, \dots, n \end{cases}$$

Having constructed Y_n , the space Y_{n-1} is obtained from Y_n by killing the homotopy of Y_n in dimension n . Then we have the following inclusions

$$X \subset Y_n \subset Y_{n-1} \subset \dots \subset Y_1$$

and we can regard it as fibrations. The fiber of $Y_q \rightarrow Y_{q-1}$ follows from homotopy exact sequence. $\square$

9.6. Computation of $\pi_4(S^3)$. The computation of $\pi_4 = \pi_4(S^3)$ is based on the fact that the homotopy group π_4 appears as the first non-trivial homology group of $K(\pi_4, 4)$. If we can fit this Eilenberg-MacLane spaces into some fibration, then its homology may be derived from Leray spectral sequence. Such fibration is obtained from Postnikov approximation.

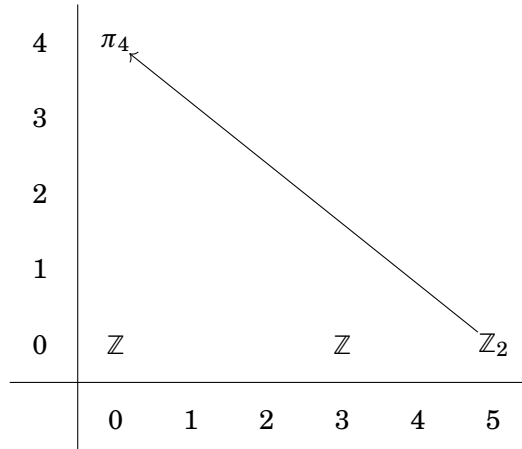
Let Y_4 be a space whose homotopy agrees with S^3 up to and including dimension 4 and vanishes in higher dimension. To get such space it suffices to kill off all homotopy groups of S^3 in dimensions ≥ 5 by attaching cells, that is

$$Y_4 = S^3 \cup e^6 \cup \dots$$

Thus Postnikov approximation gives the following fibration

$$\begin{array}{ccc} K(\pi_4, 4) & \longrightarrow & Y_4 \\ & & \downarrow \\ & & Y_3 \end{array}$$

But by definition Y_3 is $K(\mathbb{Z}, 3)$, since it is a space with the same homotopy groups as S^3 up to and including dimension 3 and vanishes in higher dimension. The E_5 -page of spectral sequence is



By basic facts about attaching cells, we know $H_4(Y_4) = H_5(Y_4) = 0$, thus the arrow shown must be an isomorphism, which implies $\pi_4(S^3) = \mathbb{Z}_2$.

Corollary 9.2 (Serre). For $n \geq 3$,

$$\pi_{n+1}(S^n) = \mathbb{Z}_2$$

Proof. This follows from $\pi_4(S^3) = \mathbb{Z}_2$ and suspension theorem. $\square$

10. SERRE'S CELEBRATED THEOREMS

In this section we mainly refer to [Hil04] and [BM08].

10.1. The Whitehead tower. The Whitehead tower is a sequence of fibrations, dual to Postnikov approximation in a certain sense, which generalizes the universal covering of a space. To be explicit, the idea of the Whitehead tower is to kill at each stage all the homotopy groups below a given dimension.

Up to homotopy the universal covering of a space X may be constructed as follows: If we use π_q to denote $\pi_q(X)$, by attaching cells to X we can kill all π_q for $q \geq 2$. Let $Y = X \cup e^3 \cup \dots$ be the space so obtained, then Y is a $K(\pi_1, 1)$ containing X as a subspace. Consider the space Ω_*^X of all paths in Y from a base point $*$ to X , the endpoint map $\Omega_*^X \rightarrow X$ is a fibration with fiber $\Omega Y = K(\pi_1, 0)$. Then apply homotopy exact sequence to

$$\begin{array}{ccc} K(\pi_1, 0) & \longrightarrow & \Omega_*^X \\ & & \downarrow \\ & & X \end{array}$$

one has

$$\cdots \rightarrow \pi_1(K(\pi_1, 0)) \rightarrow \pi_1(\Omega_*^X) \rightarrow \pi_1(X) \rightarrow \pi_0(K(\pi_1, 0)) \rightarrow 0$$

which implies $\pi_1(\Omega_*^X) = 0$, and

$$\cdots \rightarrow \pi_q(K(\pi_1, 0)) \rightarrow \pi_q(\Omega_*^X) \rightarrow \pi_q(X) \rightarrow \pi_{q-1}(K(\pi_1, 0)) \rightarrow \cdots$$

implies $\pi_q(\Omega_*^X) = \pi_q(X)$ when $q \geq 2$.

Now let's repeat this process to obtain a sequence of fibrations

$$\begin{array}{ccc} & & \vdots \\ & & \downarrow \\ K(\pi_n, n-1) & \longrightarrow & X_n \\ & & \downarrow \\ & & X_{n-1} \\ & & \downarrow \\ & & \vdots \\ & & \downarrow \\ K(\pi_1, 0) & \longrightarrow & X_1 \\ & & \downarrow \\ & & X \end{array}$$

such that

1. X_n is n -connected;
2. above dimension n the homotopy groups of X_n and X agree;
3. the fiber of $X_n \rightarrow X_{n-1}$ is $K(\pi_n, n-1)$.

This is called the Whitehead tower of X . To construct X_n from X_{n-1} , we first kill all $\pi_q(X_{n-1})$, $q \geq n+1$ by attaching cells. This gives

$$K(\pi_n, n) = X_{n-1} \cup e^{n+2} \cup \dots$$

Next let $X_n = \Omega_*^{X_{n-1}}$ be the space of all paths in $K(\pi_n, n)$ from a base point $*$ to X_{n-1} . The endpoint map gives the following fibration

$$\begin{array}{ccc} K(\pi_n, n-1) & \longrightarrow & X_n \\ & & \downarrow \\ & & X_{n-1} \end{array}$$

The homotopy exact sequence implies

$$\dots \rightarrow \pi_q(K(\pi_n, n-1)) \rightarrow \pi_q(X_n) \rightarrow \pi_q(X_{n-1}) \rightarrow \pi_{q-1}(K(\pi_n, n-1)) \rightarrow \dots$$

then if $q \neq n, n-1$, one has $\pi_q(X_n) = \pi_q(X_{n-1})$, and

$$0 \rightarrow \pi_n(X_n) \rightarrow \pi_n(X_{n-1}) \xrightarrow{\partial} \pi_{n-1}(K(\pi_n, n-1)) \rightarrow \pi_{n-1}(X_n) \rightarrow 0$$

Now it suffices to show that $\partial: \pi_n(X_{n-1}) \rightarrow \pi_{n-1}(K(\pi_n, n-1))$ is an isomorphism. Note that inclusion

$$X_{n-1} \subset K(\pi_n, n) = X_{n-1} \cup e^{n+2} \cup \dots$$

gives an isomorphism

$$\pi_n(X_{n-1}) \cong \pi_n(K(\pi_n, n))$$

By Remark 8.3 implies ∂ is exactly how $\pi_n(K(\pi_n, n))$ was identified with $\pi_{n-1}(\Omega K(\pi_n, n))$.

10.2. Serre class.

Definition 10.1 (Serre class). The Serre class $\mathcal{C}$ is a non-empty family of abelian groups such that the following axioms hold:

- I If $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$ is a short exact sequence of abelian groups, then $A \in \mathcal{C}$ if and only if $A', A'' \in \mathcal{C}$;
- II If $A, B \in \mathcal{C}$, then $A \otimes B, \text{Tor}(A, B) \in \mathcal{C}$;
- III If $A \in \mathcal{C}$, then $H_k(K(A, 1)) \in \mathcal{C}$ for every $k > 0$.

Example 10.1. The family consisting only of the zero group is a Serre class.

Example 10.2. The class of finitely generated abelian groups is a Serre class.

Example 10.3. The class of finite abelian groups is a Serre class.

Example 10.4. The class of torsion abelian groups is a Serre class.

Example 10.5. The class of abelian P -groups, where P is a family of primes, is a Serre class.

We now give our first example of a deep Serre result. Here we only need to use the first two axioms.

Theorem 10.1. Let $\pi: E \rightarrow X$ be a fibration with fiber F , X is simply-connected. Then if the homology groups of any two of E, X, F , in positive dimensions, belong to $\mathcal{C}$, so do the homology groups, in positive dimensions, of the third.

Proof. Let's prove case by case:

1. Suppose the homology groups, in positive dimensions, of X and F belong to $\mathcal{C}$, then by Axiom (I), (II) and universal coefficient theorem one has

$$E_2^{p,q} \in \mathcal{C}$$

where $(p, q) \neq (0, 0)$. Then Axiom (I) and finite convergence implies that

$$E_\infty^{p,q} \in \mathcal{C}$$

Now consider the following filtration of $H_n(E)$

$$H_n(E) = F_0 \cap H_n(E) \supset \cdots \supset F_n \cap H_n(E) \supset 0$$

It is clear $0 \in \mathcal{C}$, and $F_p \cap H_n(E)/F_{p+1} \cap H_n(E) = E_\infty^{p,n-p}$, by induction one can show $H_n(E) \in \mathcal{C}$.

2. Suppose the homology groups, in positive dimensions, of E and F belong to $\mathcal{C}$, and prove that those of X also belong to $\mathcal{C}$. If not, let m be the lowest dimension such that $H_m(X) \notin \mathcal{C}$, then $m \geq 2$, since X is simply-connected. As before $E_\infty^{p,q} \in \mathcal{C}, (p, q) \neq (0, 0), p < m$. Thus by Axiom (I), one has

$$E_r^{p,q} \in \mathcal{C}, \quad (p, q) \neq (0, 0), p < m$$

where $r \geq 2$. On the other hand, $E_2^{m,0} = H_m(X) \notin \mathcal{C}$. Now consider

$$d_2: E_2^{m,0} \rightarrow E_2^{m-2,1}$$

Then

(a) $E_3^{m,0} = \ker d_2$, since there are no maps into $E_2^{m,0}$.

(b) $\text{im } d_2$ is a subgroup of $E_2^{m-2,1}$, which implies $\text{im } d_2 \in \mathcal{C}$ by Axiom (I).

Again by Axiom (I), one has $E_3^{m,0} \notin \mathcal{C}$.

Now consider

$$d_3: E_3^{m,0} \rightarrow E_3^{m-3,2}$$

The same argument shows $E_4^{m,0} \notin \mathcal{C}$. Repeat this process one has $E_\infty^{m,0} \notin \mathcal{C}$, a contradiction to $H_m(E) \in \mathcal{C}$.

3. The proof for the remaining case is the same as above. Readers are advised to refer to [Hil04] for a more detailed proof.

□

We now exploit above Theorem 10.1 to prove a fundamental result about simply-connected spaces.

Theorem 10.2. Let X be a simply-connected space and $\mathcal{C}$ a Serre class of abelian groups. Then the homology groups $H_k(X) \in \mathcal{C}$ for $k \geq 1$ if and only if the homotopy groups $\pi_k(X) \in \mathcal{C}$ for $k \geq 1$.

First, we prove a lemma which makes essential use of Axiom (III) of Serre class.

Lemma 10.1. If $A \in \mathcal{C}$, where $\mathcal{C}$ is a Serre class, then $H_k(K(A, n)) \in \mathcal{C}$ for $n, k \geq 1$.

Proof. By Axiom (III) one has $H_k(K(A, 1)) \in \mathcal{C}$, and by Theorem 10.1 and fibration

$$\begin{array}{ccc} K(A, n-1) & \longrightarrow & PK(A, n) \\ & & \downarrow \\ & & K(A, n) \end{array}$$

one can conclude $H_k(K(A, n)) \in \mathcal{C}, n, k \geq 1$. $\square$

Proof of Theorem 10.2. Here we only prove if homology groups $H_k(X) \in \mathcal{C}$ for $k \geq 1$, then homotopy groups $\pi_k(X) \in \mathcal{C}$ for $k \geq 1$. Consider the Whitehead tower of X , and it begins from X_2 , since X is simply-connected, that is

$$\begin{array}{ccc} K(\pi_2, 1) & \longrightarrow & X_2 \\ & & \downarrow \\ & & X \end{array}$$

Hurewicz theorem implies $\pi_2 = H_2(X)$, thus homology groups of $K(\pi_2, 1)$ in positive dimensions in $\mathcal{C}$, then those of X_2 are also in $\mathcal{C}$ by Theorem 10.1. However X_2 is 2-connected, and $\pi_3(X_2) = \pi_3$, then Hurewicz theorem implies $\pi_3 \in \mathcal{C}$. Now consider

$$\begin{array}{ccc} K(\pi_3, 2) & \longrightarrow & X_3 \\ & & \downarrow \\ & & X_2 \end{array}$$

The same argument shows homology groups of X_3 in positive dimensions in $\mathcal{C}$, and thus $\pi_4 \in \mathcal{C}$. By induction one can show all $\pi_k \in \mathcal{C}, k \geq 1$. $\square$

Corollary 10.1 (Serre). All homotopy groups of S^n are finitely generated.

10.3. Serre's torsion theorem.

Theorem 10.3 (Serre). The homotopy groups of an odd sphere are torsion except in dimension n ; those of an even sphere are torsion except in dimensions n and $2n - 1$.

Proof. The essential facts to be used in the proof are the following:

- (a) In the Whitehead tower of any space X , $\pi_{q+1}(X) = \pi_{q+1}(X_q) = H_{q+1}(X_q)$, thus

$$\pi_{q+1}(X) \otimes \mathbb{Q} = H_{q+1}(X_q; \mathbb{Q})$$

- (b) The rational cohomology ring of $K(\pi, n)$ is trivial for a torsion finitely generated abelian group π , and is free on one generator of dimension n for $\pi = \mathbb{Z}$.

Since S^n is $(n-1)$ -connected and $\pi_n(S^n) = \mathbb{Z}$, then Whitehead tower begins with

$$\begin{array}{ccc} K(\mathbb{Z}, n-1) & \longrightarrow & X_n \\ & & \downarrow \\ & & S^n \end{array}$$

For convenience we use π_q to denote $\pi_q(S^n)$.

1. Assume that n is odd and $n \geq 3$. The rational cohomology of $K(\mathbb{Z}, n-1)$ is a polynomial algebra on one generator of dimension $n-1$ and the E_2 -page of the above fibration is

$$\begin{array}{ccccc} & & & & \\ & & & & \\ 2(n-1) & & \mathbb{Q} & & \mathbb{Q} \\ & & \searrow & & \\ & & & & \mathbb{Q} \\ n-1 & & \mathbb{Q} & & \mathbb{Q} \\ & & \searrow & & \\ & & & & \mathbb{Q} \\ & & & & \\ & & & & n \end{array}$$

The bottom arrow is an isomorphism since $H_{n-1}(X_n; \mathbb{Q}) = 0$, which implies $H^{n-1}(X_n; \mathbb{Q}) = 0$; the other arrows are all isomorphisms by the product structure. From this we can see X_n has trivial rational cohomology, hence trivial rational homology. By (a) one can conclude π_{n+1} is torsion. Now consider

$$\begin{array}{ccc} K(\pi_{n+1}, n) & \longrightarrow & X_{n+1} \\ & & \downarrow \\ & & X_n \end{array}$$

Since both X_n and $K(\pi_{n+1}, n)$ have trivial rational cohomology, X_{n+1} does as well; thus, X_{n+1} has trivial rational homology, and by the same reason one has π_{n+2} is torsion. By induction one has $q \geq n+1$, π_q is torsion.

2. If n is even, then rational cohomology of $K(\mathbb{Z}, n-1)$ is an exterior algebra and the E_2 -page of homology spectral sequence is

$$\begin{array}{ccccc} & & & & \\ & & & & \\ n-1 & & \mathbb{Q} & & \mathbb{Q} \\ & & \nwarrow & & \\ & & & & \mathbb{Q} \\ & & & & \\ & & & & n \end{array}$$

The arrow shown is an isomorphism, since X_n is n -connected, so

$$H_k(X_n; \mathbb{Q}) = \begin{cases} \mathbb{Q}, & k = 0, 2n-1 \\ 0, & \text{otherwise} \end{cases}$$

- I Suppose $n > 2$, then $n + 1 < 2n - 1$. By (a), one has $\pi_{n+1} = H_{n+1}(X_n)$ is torsion since $H_{n+1}(X_n; \mathbb{Q}) = 0$. Thus $H_*(K(\pi_{n+1}, n); \mathbb{Q})$ is trivial, from the following fibration

$$\begin{array}{ccc} K(\pi_{n+1}, n) & \longrightarrow & X_{n+1} \\ & & \downarrow \\ & & X_n \end{array}$$

one can conclude X_{n+1} has the same rational homology as X_n , which implies π_{n+2} is also torsion. This argument still holds, until X_{2n-2} has the same rational homology as X_{2n-3} (which relies on π_{2n-2} is torsion). Thus $H_{2n-1}(X_{2n-2}; \mathbb{Q}) = H_{2n-1}(X_n; \mathbb{Q}) = \mathbb{Q}$, which implies π_{2n-1} is not torsion⁶.

- II Now suppose $n \geq 2$, by (b) one has the rational cohomology ring $H^*(K(\pi_{2n-1}, 2n-2); \mathbb{Q})$ is a polynomial algebra on one generator, so the E_2 -page of fibration $K(\pi_{2n-1}, 2n-2) \rightarrow X_{2n-1} \rightarrow X_{2n-2}$ is

$$\begin{array}{ccccc} & & & & 2n-1 \\ & & & & \mathbb{Q} \\ 4n-4 & & \mathbb{Q} & \longleftarrow & \mathbb{Q} \\ & & & & \downarrow \\ 2n-2 & & \mathbb{Q} & \longleftarrow & \mathbb{Q} \\ & & & & \downarrow \\ & & \mathbb{Q} & & \mathbb{Q} \end{array}$$

Since $H_{2n-1}(X_{2n-1}) = 0$, then the arrows shown must all be isomorphisms. It follows the rational cohomology of X_q is trivial except in dimension 0 for all $q \geq 2n-1$. In particular, one has

$$\pi_{q+1} \otimes \mathbb{Q} \cong H_{q+1}(X_q; \mathbb{Q}) = 0$$

which implies π_q is torsion for $q > 2n-1$.

□

⁶In fact, π_{2n-1} has one infinite cyclic generator and possibly some torsion generators.

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